

Distribution-independent SQ learning does not imply low dimension complexity

Research working draft

September 10, 2026

Third research draft

Abstract

If every product distribution on a finite truth table has a monochromatic rectangle of mass at least $1/\rho$, one distribution-independent SQ learner uses $O(\rho(\log |H| + \log(1/\epsilon)))$ queries of tolerance $\Omega(\epsilon/(\rho \log(1/\epsilon)))$ to achieve expected error at most ϵ against every valid adaptive oracle. This rectangle theorem has two consequences. First, the random monotone incidence flips of Hatami et al., on $2M$ points, are SQ-learnable with $O(\log M)$ queries at fixed accuracy while their dimension is $\Omega(M^{1/3}/\log M)$, their exact probabilistic dimension is $\Omega(M^{1/3}/\log^2 M)$, and their averaged probabilistic dimension is $\Omega_\eta(M^{1/3}/\log M)$ for every fixed $\eta < 1/2$. Second, a cap implementation for fixed-dimensional templates is polynomial-time in the supplied table, admits a deterministic version, and yields a proper learner with $O(\log M + 1/\epsilon + \log(1/\epsilon))$ queries. We prove that the rectangle condition is sufficient but not necessary, quantify marginal adaptation, and refute a source-description-length repair when computation is free. The high-dimension selection is not efficiently explicit. The specified Gaussian-initialized plain-SGD question remains open. Exact oracle and finite-coverage CPU audits were executed; the pinned Lean sources remain uncompiled.

1 Introduction

Feldman, Kamath, and Srebro ask whether distribution-independent statistical-query learning forces a common low-dimensional linear representation [1]. We answer their SQ question negatively, including its polynomial and probabilistic-dimension variants. The distinction is between *distribution independence*, which asks for one algorithm valid under every marginal, and *distribution adaptation*, which permits that algorithm to ask where the unknown marginal places mass.

A linear representation must fix every example-hypothesis sign at once. A learner can instead query marginal masses, label regions on which its surviving hypotheses agree, and use disagreements to eliminate hypotheses. A product-distribution rectangle condition is exactly what a marginal-free mixture can cover pointwise, by finite minimax. The new step is to turn that mixture into an adversarial-oracle learner whose progress is the product of the two rectangle masses. A bounded-drift argument gives a worst-case query budget, not just finite expected termination. The random incidence flips of Hatami et al. [2] have large rectangles and large sign-rank, so these two resources separate even for one algorithm serving all marginals.

Contributions.

- If $\text{rect}(H) \geq 1/\rho$, RECTIFY uses $O(\rho(\log |H| + \log(1/\epsilon)))$ queries at tolerance $\Omega(\epsilon/(\rho \log(1/\epsilon)))$ to achieve expected error at most ϵ against every valid adaptive oracle. Applied to the incidence

flips on $2M$ points, it refutes all three polynomial dimension implications, including $\text{dc}_{1/2}$ and $\text{dc}^{C\epsilon}$.

- A fixed-dimensional strict template with no negative 2×2 admits a pointwise cap sampler and a finite-bit implementation polynomial in the supplied table. A finite net also gives a deterministic learner. For the flip family a proper completion has $O(\log M + 1/\epsilon + \log(1/\epsilon))$ queries at the same tolerance order.
- We prove the all-product bound $\text{rect}(\text{PG}(2, q)) = \Theta(1/q)$, exhibit SQ-easy cube halfspaces with $\text{rect} \leq e^{-n/4}$, quantify necessary marginal adaptation under a small-query budget, and show that source-description length alone does not repair the implication when computation is uncharged.

Imported results and attribution. The high-sign-rank construction is due to Hatami et al. [2], and its counting uses Warren’s strict-sign-pattern theorem [4, 5]. The homogeneous-rectangle theorem [3] is used only for the sharper constant 2^{-15} ; the self-contained cap proof already gives a constant sufficient for the separation. The minimax rectangle-mixture device also appears in the average-margin proof of Hatami et al. [2, Theorem 3.1]; we do not claim that device as new. The new learning argument is its adaptive peeling/elimination analysis, its uniform query-budget proof, and the proper completion. The probabilistic lower bounds use the actual target-dependent marginal quantifiers, rather than uniform approximate sign-rank. Feldman’s characterization is invoked only for the separate comparison in Section 6, not for the main learner. We additionally use real-closed-field decidability in the program-length counterexample and state that imported algorithmic fact explicitly. The weakly polynomial convex-program implementation needed for caps is proved in Appendix B.

1.1 Related work

Forster’s spectral approach connects sign-rank with normalized vector representations [7]. Hatami et al. compare VC dimension, average margin and monochromatic-rectangle ratio, and construct matrices that separate these parameters from sign-rank [2]. Their geometric and counting ingredients motivate the present learner; they do not themselves provide its unknown-marginal oracle analysis.

Feldman’s 2017 characterization concerns general randomized SQ search complexity, and his finite-field line example separates distribution-specific from distribution-independent SQ learning [6]. Our class lies on the easy side of distribution-independent SQ learning, so that older separation does not settle the question addressed here. Approximate radial-isotropy algorithms have been developed independently, including strongly polynomial methods [12]. We use their geometric object, but only claim weakly polynomial computation and include a direct fixed-dimensional implementation proof. The engineered differentiable-learning simulations of Abbe et al. [8] do not preserve the Gaussian-initialized bias-free architecture in the remaining SGD question. Recent topological sign-rank bounds and upper bounds for Hamming-distance matrices are discussed in Section 11 [10, 11].

2 Definitions and quantifiers

Let X be nonempty and finite, and let the nonempty indexed class $H = \{h_1, \dots, h_K\} \subseteq \{-1, +1\}^X$ be finite. Duplicate rows cause no difficulty. Set $n = \log_2 |X|$. For a distribution D on X and

Boolean predictor g , define

$$L_{D,h}(g) = \sum_x D(x) \mathbf{1}_{g(x) \neq h(x)}, \quad \mathbb{E}_D[hg] = 1 - 2L_{D,h}(g).$$

A query is any $q : X \times \{-1, +1\} \rightarrow [-1, 1]$. At tolerance $\tau > 0$, a valid oracle returns any number within τ of $\mathbb{E}_D q(x, h(x))$. Its choices may depend on the query history. A randomized learner must satisfy

$$\forall D \forall h \in H \forall \text{ valid oracle policies } O, \quad \mathbb{E}_\omega L_{D,h}(A_\omega^O) \leq \epsilon. \quad (1)$$

The private seed is sampled independently of the instance. A deterministic algorithm is a real-response decision tree; a randomized algorithm is a distribution over such trees. Query count is a worst-case bound. Computation and randomness length are not charged. All random variables used here are measurable; our learner has finitely many discrete states and uses only threshold comparisons of answers.

The dimension $\text{dc}(H)$ is the least $d \geq 0$ admitting $\phi : X \rightarrow \mathbb{R}^d$ such that every h has a vector w_h with

$$h(x) \langle w_h, \phi(x) \rangle > 0 \quad (x \in X). \quad (2)$$

It equals the sign-rank of the truth table, since a rank factorization is exactly such a representation. It is finite by $\phi(x) = e_x$.

For the probabilistic variants we use $\text{sign}(0) = +1$, without changing the strict definition above. Appendix A verifies that both lower bounds also hold under the opposite tie rule or when zeros are counted as errors. The quantity $\text{dc}_{1/2}(H)$ is the least d for which one distribution over embeddings satisfies

$$\forall D, h, \quad \Pr_\phi[\exists w : L_{D,h}(\text{sign}\langle w, \phi \rangle) = 0] \geq 1/2.$$

The quantity $\text{dc}^\eta(H)$ instead requires

$$\forall D, h, \quad \mathbb{E}_\phi \inf_w L_{D,h}(\text{sign}\langle w, \phi \rangle) \leq \eta.$$

The embedding law is independent of both D and h in each definition.

A monochromatic rectangle is $S \times T \subseteq X \times H$ with $h(x) = c$ for all $x \in S, h \in T$, for some $c \in \{-1, +1\}$. Define

$$\text{rect}(H) = \inf_{\mu \in \Delta(X), \nu \in \Delta(H)} \max_{S \times T \text{ monochromatic}} \mu(S) \nu(T).$$

All rectangles and simplices are finite-dimensional. Restricting to any nonempty submatrix cannot decrease rect , by extending its two distributions by zero. In particular, a known rational $r = 1/\rho \leq \text{rect}(H)$ remains valid throughout the algorithm.

3 The rectangle learner

The earlier barrier and transcript results retain numbers 1–5 in Appendix G. The main new results start at 9.

Theorem 9 (Distribution-independent rectangle learner). *Suppose $\rho \geq 1$ and $\text{rect}(H) \geq 1/\rho$. For every $0 < \epsilon < 1/4$, one randomized algorithm satisfies (1) with*

$$m = O(\rho(\log K + \log(1/\epsilon))), \quad \tau = \Omega\left(\frac{\epsilon}{\rho \log(1/\epsilon)}\right).$$

Query count is a worst-case bound. The learner is explicit from the truth table and a valid lower bound on $\text{rect}(H)$; the general implementation may solve exponential-size linear programs.

Lemma 9.1 (Exact parameters). *The theorem holds with*

$$r = 1/\rho, \quad P = 1 + \lceil 4\rho \ln(8/\epsilon) \rceil, \quad \tau = \epsilon/(24P), \quad (3)$$

$$B = \ln K + \ln(8/\epsilon) + 1, \quad R = \lceil 8\rho(B + \ln(2/\epsilon)) \rceil, \quad m = 3R + 1. \quad (4)$$

Lemma 10 (A marginal-free rectangle mixture). *Fix nonempty $U \subseteq X$, nonempty $V \subseteq H$, and a distribution ν on V . There is a distribution λ over monochromatic rectangles $S_R \times T_R \subseteq U \times V$ such that*

$$\forall x \in U, \quad \mathbb{E}_{R \sim \lambda}[\mathbf{1}_{S_R}(x)\nu(T_R)] \geq r. \quad (5)$$

This distribution can be computed from (H, U, V, ν) without a query about D .

Proof. This minimax device is also used in Hatami et al. [2, Theorem 3.1]; we include the proof and the rational feasibility detail. For each rectangle let $z_R \in \mathbb{R}^U$ have coordinates $z_R(x) = \mathbf{1}_{S_R}(x)\nu(T_R)$. The rectangle assumption gives

$$\forall \mu \in \Delta(U), \quad \max_R \langle \mu, z_R \rangle \geq r.$$

Let C be the convex hull of the finitely many z_R and let $Q = r\mathbf{1} + \mathbb{R}_{\geq 0}^U$. Suppose $C \cap Q = \emptyset$. The distance between compact C and closed Q has a positive attained minimum: a minimizing sequence has bounded Q -components and convergent subsequences. Let (c, z) attain it and put $v = z - c \neq 0$. Differentiating squared distance along segments to any $c' \in C, z' \in Q$ gives

$$\langle v, c' - c \rangle \leq 0, \quad \langle v, z' - z \rangle \geq 0.$$

Taking $z' = z + te_x$ gives $v_x \geq 0$. Hence

$$\max_{c' \in C} \langle v, c' \rangle \leq \langle v, c \rangle < \langle v, z \rangle = \min_{z' \in Q} \langle v, z' \rangle = r \sum_x v_x.$$

Normalize v to a distribution to contradict the assumption. A point in $C \cap Q$ supplies λ . Enumerating all rectangles and solving these finite linear inequalities computes it. For rational ν, r , the feasible polytope has rational coefficients and admits a rational feasible point: Gaussian elimination on a minimal face yields such a point. Thus no oracle about D enters this computation. $\square$

RECTIFY: rectangle peeling and elimination.

Initialize $U = X$, $V = H$, and an empty partial predictor. At every state let ν be uniform on V . Use the constants in (3)–(4). Repeat the following for at most R rounds.

1. If $U = \emptyset$, stop. Query $\mathbf{1}_U(x)$ and call its answer $\hat{u}$. If $\hat{u} \leq \epsilon/4$, stop and fill U with $+1$.
2. Compute the distribution in Lemma 10. Draw a monochromatic rectangle $S \times T$ of color c . Query $\mathbf{1}_S(x)$, with answer $\hat{s}$. If $\hat{s} < (r/2)\hat{u}$, reject this proposal and go to the next round.
3. Query $\mathbf{1}_S(x)\mathbf{1}_{y \neq c}$, with answer $\hat{w}$. If $\hat{w} \leq 2\tau$, label S by c and replace U by $U \setminus S$. Otherwise replace V by $V \setminus T$.
4. At the round limit, stop and fill any remaining points with $+1$.

Every accepted peel removes a subset of the current U , so peeled sets are disjoint. An elimination does not change U . This general implementation requires no candidate test, conditional-label oracle, isotropic transform, or subspace restriction. Section 7 supplies the separate cap implementation.

Proof of Theorem 9. Fix D, h^* and any valid oracle policy. First,

$$\tau \leq \epsilon/8, \quad \tau \leq r\epsilon/64. \quad (6)$$

Indeed $P \geq 4\rho \ln(8/\epsilon) > 4\rho \ln 32$, and $24 \cdot 4 \ln 32 > 64$. At a nonstopping round, write

$$u = D(U) > \epsilon/4 - \tau \geq \epsilon/8, \quad a = D(S)/u, \quad b = \nu(T), \quad w = D(S \cap \{h^* \neq c\}).$$

Thus $\tau/u \leq r/8$. If the proposal is accepted, validity implies

$$D(S) \geq (r/2)u - (1 + r/2)\tau \geq (5r/16)u \geq (r/4)u.$$

Conversely, every rectangle with $a \geq 3r/4$ is accepted, since

$$\hat{s} - (r/2)\hat{u} \geq (r/4)u - (1 + r/2)\tau \geq (r/16)u > 0.$$

These inequalities hold at the extreme valid oracle answers, not only for exact answers.

If $h^* \in T$, monochromaticity gives $w = 0$, hence $\hat{w} \leq \tau < 2\tau$. Therefore the algorithm never eliminates the target. In particular, V remains nonempty. On an elimination, its normalized target weight increases by a factor $1/(1-b)$. Because $-\ln(1-b) \geq b$ for $0 \leq b < 1$, summing gives

$$\sum_{\text{eliminations}} b \leq \ln K. \quad (7)$$

For completeness, $\ln t \leq t-1$ for $t > 0$ follows by minimizing $t-1-\ln t$, whose derivative is $1-1/t$. Taking $t = 1-b$ proves the displayed inequality. The target weight starts at $1/K$ and never exceeds one.

Each peel multiplies u by at most $1-r/4$. Immediately before every peel, $u \geq \epsilon/8$. Before the j th peel,

$$\epsilon/8 \leq (1-r/4)^{j-1} \leq e^{-(j-1)r/4}.$$

Hence there are at most P peels. Their combined wrong-color mass is at most $3P\tau = \epsilon/8$, since a peeling answer implies $w \leq 3\tau$. At a normal stop the residual mass is at most $\epsilon/4 + \tau$, or zero. The error at such a stop is at most $\epsilon/2$.

It remains to bound the probability of reaching the round limit. Define a round's progress by $G = a$ for a peel, $G = b$ for an elimination, and $G = 0$ for a rejected proposal. Conditional on the pre-proposal history, $0 \leq G \leq 1$. For every accepted rectangle, either action has $G \geq ab$. All rectangles with $a \geq 3r/4$ are accepted. Averaging (5) under D restricted to U gives $\mathbb{E}[ab] \geq r$. Consequently,

$$\mathbb{E}[G \mid \text{history}] \geq \mathbb{E}[ab] - \mathbb{E}[ab \mathbf{1}_{a < 3r/4}] \geq r - (3r/4)\mathbb{E}b \geq r/4. \quad (8)$$

This uses no favorable distributional assumption and holds for either oracle-induced action on every accepted rectangle.

On any trajectory of nonstopping rounds, the sum of all peeling a values except the last is at most the logarithm of the reciprocal residual mass immediately before the last peel. Indeed $a \leq -\ln(1-a)$ and the residual ratios telescope. This sum is at most $\ln(8/\epsilon)$; the last a is at most one. If there are no peels the sum is zero. Together with (7), this proves

$$\sum G \leq B. \quad (9)$$

There is no logarithm of zero: a peel with $a = 1$ must be the last peel.

After any normal termination, extend the sequence by setting $G = 1$. Let $p = r/4$. For $0 \leq g \leq 1$, convexity places e^{-g} below the chord between its endpoint values:

$$e^{-g} \leq 1 - (1 - e^{-1})g.$$

Since $1 - e^{-1} > 1/2$, conditional expectation and (8) imply

$$\mathbb{E}[e^{-G} \mid \text{history}] \leq e^{-p/2}.$$

Induction gives $\mathbb{E}e^{-\sum_{t=1}^R G_t} \leq e^{-pR/2}$. On the event of no normal termination by the budget, (9) holds. Markov's inequality therefore gives

$$\Pr[\text{budget termination}] \leq e^{B-pR/2} \leq \epsilon/2.$$

A point that becomes empty after the final peel already has the normal-stop error bound; counting it as budget termination only weakens this estimate. Finally, error is always at most one, so expected error is at most $\epsilon/2 + \epsilon/2 = \epsilon$. There are at most three queries per round and at most one extra stopping check. The big- O and Ω bounds follow directly from the ceilings and $\rho \geq 1$, $\epsilon < 1/4$. $\square$

Domain-size consequence of Theorem 9. The dependence on $\log K$ can always be bounded in terms of the domain and ρ . If H shatters $v \geq 1$ points, put the uniform marginal on those points and the uniform prior on one representative of each of their 2^v label patterns. A nonempty monochromatic rectangle using s of the points has mass at most $s/(v2^s) \leq 1/(2v)$, because $s \leq 2^{s-1}$ for every positive integer s (by induction). Therefore

$$\text{VC}(H) \leq \rho/2.$$

After removing duplicate rows, the finite counting inequality $K \leq \sum_{i=0}^v \binom{|X|}{i}$ follows by induction on $|X|$: split restrictions on the last point into their union A and intersection B . Their sizes sum to K , the VC dimension of A is at most v , and that of B is at most $v-1$, since shattering a set in B also permits either label on the last point. Pascal's identity completes the induction, with the empty domain as base case. For $v = 0$, all rows are identical and $K = 1$. For $1 \leq v \leq Q = |X|$, setting $t = v/Q \leq 1$ gives

$$t^v \sum_{i=0}^v \binom{Q}{i} \leq (1+t)^Q \leq e^v, \quad \ln K \leq v \ln(eQ/v) \leq (\rho/2) \ln(eQ).$$

Consequently the same learner, with no statistical-dimension conversion, has

$$\boxed{m = O(\rho^2 \ln(e|X|) + \rho \ln(1/\epsilon)), \quad \tau = \Omega(\epsilon/(\rho \ln(1/\epsilon)))}.$$

For bounded ρ , this is logarithmic in the domain size at fixed accuracy. The argument is a proved consequence, not a conjectured converse characterizing SQ complexity.

4 The incidence-flip family

Let $N \geq 2$, $M = N^3$, and $K = 2M$. Points are $p = (x, y) \in [N] \times [2N^2]$. Rows are affine lines $\ell = (a, b) \in [N] \times [2N^2]$. Put $F_{\ell p} = -1$ exactly when $y = ax + b$. A monotone flip matrix A changes any subset of these -1 entries to $+1$ and leaves all other entries unchanged. In the random family the choices are independent fair bits. Denote the row class by H_A .

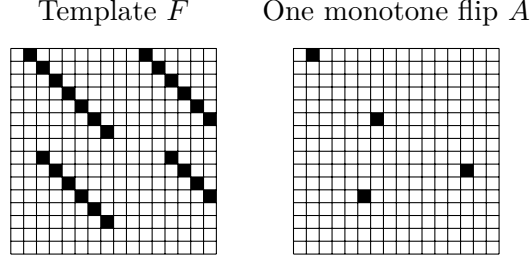


Figure 1: The $N = 2$ template and the seeded flip used in the CPU tests. Black cells are negative, white cells positive. Rows index lines; columns index points. Only black cells may change. Larger N , not this small example, yields the asymptotic counting separation.

Theorem 11 (Construction and ordinary dimension). *Every monotone flip A satisfies $\text{VC}(A) \leq 2$ and $\text{rect}(A) \geq 2^{-15}$. The template has $\text{signrank}(F) \leq 6$ and exactly*

$$I = 2N^4 - \left(\frac{N(N+1)}{2} \right)^2 \geq N^4 \quad (10)$$

independent flip positions. Set $B_0 = \log_2(4eK)$ and $s_0 = \lfloor I/(4KB_0) \rfloor$. A random A satisfies

$$\Pr[\text{dc}(H_A) \leq s_0] \leq 2^{-I/2}.$$

Every member also satisfies $\text{dc}(H_A) \leq 2N + 1$. In particular most matrices have $\text{dc}(H_A) = \Omega(N/\log M)$. All members have the learner of Theorem 9 with $\rho = 2^{15}$. Independently of the sharp rectangle input, the self-contained cap proof in Appendix F supplies $\rho = 2^{18}$ and the same asymptotic learning bound.

Proof. Define

$$u_p = (x^2, xy, y^2, x, y, 1), \quad v_\ell = (a^2, -2a, 1, 2ab, -2b, b^2 - 1/2).$$

Their dot product is $(ax + b - y)^2 - 1/2$. It is negative on incidences and positive on nonincidences because all coordinates are integers. Thus the six-dimensional strict factorization proves $\text{signrank}(F) \leq 6$. Both vectors are nonzero and may be normalized to unit vectors. Every row of A has at most N negative points. Enumerate all domain points by $t \in [K]$ and use $(1, t, \dots, t^{2N})$. For a row with negative index set S , the polynomial $\prod_{a \in S} (t - a)^2 - 1/2$ is negative exactly on S and positive at every other integer domain point. The empty product is one. This proves $\text{dc}(H_A) \leq 2N + 1$ for every member.

For fixed a, x , exactly $2N^2 - ax$ values of $b \in [2N^2]$ give an incidence with $y \in [2N^2]$. Summing proves (10). The lower bound uses $(N+1)/2 \leq N$. If two distinct affine lines share points (x, y) and (x', y') , subtraction gives $(a - c)(x - x') = 0$. If $a = c$, their shared point forces equal intercepts, a contradiction. Otherwise $x = x'$ and then $y = y'$. Hence F , and therefore A , has no all-negative 2×2 . Shattering three columns would require one row negative on all three and a distinct row negative on two of them, which creates such a submatrix. Therefore $\text{VC}(A) \leq 2$.

The homogeneous-rectangle theorem of Alon et al. [3, Theorem 1.3], in the formulation checked in Hatami et al. [2, Theorem 1.9], states that an $a \times b$ sign matrix of sign-rank at most d contains a monochromatic rectangle with each side at least the corresponding side length divided by 2^{d+1} . For rational product weights, repeat rows and columns according to common-denominator multiplicities. Projecting the resulting rectangle back to the original matrix preserves its signs and can only

increase its weighted masses. For arbitrary weights, approximate by rationals; the maximum over finitely many rectangle masses is continuous. It follows that

$$\text{rect}(F) \geq 2^{-2d-2}. \quad (11)$$

Fix a product distribution and such a rectangle of F . If its color is positive, it survives every monotone flip. If its color is negative, one side is a singleton, by the absence of a negative 2×2 . Partition the other side according to its sign in A ; one part retains at least half its mass. Thus

$$\text{signrank}(F) \leq d, \quad F \text{ has no negative } 2 \times 2 \implies \text{rect}(A) \geq 2^{-(2d+3)}. \quad (12)$$

For $d = 6$ this is 2^{-15} . This reduction establishes exactly which extra structural hypothesis is needed.

For the counting bound, the theorem of Warren [4] says that $s \geq v$ real polynomials of degree at most k in v variables have at most $(4eks/v)^v$ strict sign patterns; see Alon et al. [5, Theorem 21 and Lemma 22]. A rank-at-most- d $K \times K$ matrix factors as a $K \times d$ matrix times a $d \times K$ matrix. Its K^2 entries are degree-two polynomials in $2Kd$ variables. Therefore, for $1 \leq d \leq K/2$,

$$\#\{B \in \{-1, +1\}^{K \times K} : \text{signrank}(B) \leq d\} \leq (4eK/d)^{2Kd} \leq 2^{2KB_0}. \quad (13)$$

The random family has exactly 2^I equiprobable matrices. Taking $d = s_0$ bounds the probability by $2^{2Ks_0B_0-I} \leq 2^{-I/2}$. The range $s_0 \leq K/2$ follows from $I \leq K^2$ and $B_0 > 1$; when $s_0 = 0$ the event is empty. Finally $I \geq N^4$ and $K = 2N^3$ give the asymptotic lower bound. Apply Theorem 9 for the learner. $\square$

The restriction in (12) cannot be omitted. Every sign matrix is a monotone flip of the all-negative, rank-one matrix. More quantitatively, let W be a $k \times k$ Walsh matrix, k a power of two. Its orthogonality gives $\|W\|_{\text{op}} = \sqrt{k}$. A monochromatic rectangle of side sizes a, b satisfies

$$ab = |\mathbf{1}_S^T W \mathbf{1}_T| \leq \sqrt{k} \sqrt{ab},$$

so its uniform product mass is at most $1/k$. For $k > 32$, this contradicts the asserted unrestricted bound $2^{-(2 \cdot 1 + 3)}$.

5 Probabilistic dimension under the actual definitions

Theorem 12 (Approximate counting and exact probabilistic dimension). *Fix $0 \leq \eta < 1/2$ and write*

$$h_2(\eta) = -\eta \log_2 \eta - (1 - \eta) \log_2 (1 - \eta), \quad c_\eta = 1 - h_2(\eta) > 0, \quad s_\eta = \left\lfloor \frac{c_\eta I}{4KB_0} \right\rfloor,$$

with $0 \log 0 = 0$. For the random incidence-flip matrix,

$$\Pr[\text{dc}^\eta(H_A) < s_\eta] \leq 2^{-c_\eta I/2}. \quad (14)$$

For every finite class, independently of this construction,

$$\text{dc}(H) \leq (\lceil \log_2 K \rceil + 1) \text{dc}_{1/2}(H) + 1. \quad (15)$$

Consequently most A have $\text{dc}_{1/2}(H_A) = \Omega(N/\log^2 M)$ and $\text{dc}^\eta(H_A) = \Omega_\eta(N/\log M)$.

Proof. Let J_ℓ be the set of template incidences in row ℓ , with $k_\ell = |J_\ell|$. For each nonempty J_ℓ , take D_ℓ uniform on it. If a law on d -dimensional embeddings witnesses $\text{dc}^\eta(H_A) \leq d$, then

$$\mathbb{E}_\phi \sum_{\ell: k_\ell > 0} \frac{k_\ell}{I} \min_w L_{D_\ell, h_\ell}(\text{sign}\langle w, \phi \rangle) \leq \eta.$$

For a fixed embedding, only finitely many Boolean predictions are possible, so each infimum is attained as a minimum of finitely many loss values. There exists one embedding with the displayed weighted sum at most η . Otherwise its positive excess over η would have strictly positive expectation. Choose one minimizing weight vector for each row. Their prediction matrix makes at most ηI mistakes on the incidence positions.

With $\text{sign}(0) = +1$, this prediction matrix has strict sign-rank at most $d + 1$, not necessarily at most d . Add the constant feature one. In each row add a strictly positive bias smaller than the absolute value of every negative score in that row, if there are negative scores. This retains all signs and turns every zero into a positive score. Finiteness makes the bias choice possible. Rows with no incidence may use any weight vector.

For any fixed sign matrix B , the number of possible incidence flip patterns within ηI mistakes is at most

$$\sum_{j=0}^{\lfloor \eta I \rfloor} \binom{I}{j} \leq 2^{h_2(\eta)I}.$$

For $0 < \eta < 1/2$, this follows from

$$1 \geq \sum_{j \leq \eta I} \binom{I}{j} \eta^j (1 - \eta)^{I-j} \geq 2^{-h_2(\eta)I} \sum_{j \leq \eta I} \binom{I}{j};$$

the second inequality uses $j \leq \eta I$ and $\eta/(1 - \eta) < 1$. For $\eta = 0$ the sum is one. Combining with (13), the probability that any rank-at-most- s_η sign matrix makes at most ηI incidence errors is at most

$$2^{2Ks_\eta B_0 + h_2(\eta)I - I} \leq 2^{-c_\eta I/2}.$$

An embedding of dimension below s_η would produce just such a strict sign matrix after adding the constant coordinate. If $s_\eta = 0$, the event in question is empty. This proves (14).

For (15), choose a full-support distribution on X . Zero loss under this distribution means correct labels at every point. Draw $t = \lceil \log_2 K \rceil + 1$ independent embeddings from a law witnessing $\text{dc}_{1/2}(H) = d$. Each fixed target fails in all t embeddings with probability at most 2^{-t} . A union bound gives probability at most $K2^{-t} < 1$ that some target fails everywhere. Hence one collection works for all targets. Concatenate its t embeddings; each row selects a successful block. Adding one constant coordinate as above makes the representation strict. This proves (15). Combine with Theorem 11. If simultaneous bounds are desired, intersect its event with (14); failure probability is at most $2^{-I/2} + 2^{-c_\eta I/2}$. $\square$

For fixed $\epsilon > 0$, $\rho = 2^{15}$ gives $m = O(\log M) = O(n)$ and $1/\tau = O_\epsilon(1)$, whereas the displayed dimensions are exponential in n , up to logarithmic factors. Thus forms (a), (b), and (c) fail for dc and $\text{dc}_{1/2}$. For any proposed fixed constant $C > 0$ in $\text{dc}^{C\epsilon}$, fix $\epsilon < \min\{1/4, 1/(2C)\}$ and apply Theorem 12 at $\eta = C\epsilon$. This refutes the same three polynomial forms for that variant. The assertion $m/\tau^2 = O(\log n)$ is not used and is not implied: the proved bound is $O_\epsilon(n)$.

6 A direct check through randomized statistical dimension

We give the proposed dimension route separately. It is not needed for Theorem 9. For the search task $\mathcal{Z}_\eta$, an instance is a realizable labeled distribution $P_{D,h}$; an allowed solution is a predictor g with loss at most η . Fix any reference distribution P_0 on $Z = X \times \{-1, +1\}$, not necessarily realizable or fair-labeled. For a law $\mathcal{P}$ on predictors, put

$$\mathcal{B}(\mathcal{P}) = \{j : \Pr_{g \sim \mathcal{P}}[g \text{ is } \eta\text{-good for } j] < 1/4\}.$$

A query distribution covers this bad set at level $1/d$ when every j in it is distinguished from P_0 by more than κ with probability at least $1/d$. Taking the best $\mathcal{P}$ and query law, and the worst P_0 , is the covering formulation of Feldman's $\text{RSD}_{\kappa_1}(\mathcal{Z}_\eta, \kappa, 1/4)$.

Theorem 13 (Rectangle bound for randomized statistical dimension). *Let $L = \lceil \rho \ln(4/\eta) \rceil$ and $\kappa = \eta/(4L)$, where $0 < \eta < 1/4$. Then*

$$\text{RSD}_{\kappa_1}(\mathcal{Z}_\eta, \kappa, 1/4) \leq 4L.$$

This statement allows arbitrary target-dependent marginals and any reference P_0 .

Proof. Fix a finitely supported prior π on instances. It induces a prior ν on hypotheses. Let $\mathcal{Q}$ be the finite query family $q_{S,c}(x, y) = \mathbf{1}_S(x)\mathbf{1}_{y \neq c}$. Set

$$\beta = \max_{q \in \mathcal{Q}} \Pr_{j \sim \pi}[|P_j q - P_0 q| > \kappa].$$

Peel L rectangles using the prior-averaged example marginal restricted to the current unpeeled set U , and the unchanged hypothesis prior ν . Empty residual mass permits immediate completion. The rectangle condition gives both

$$\nu(T) \geq r, \quad \mathbb{E}_j D_j(S) \geq r \mathbb{E}_j D_j(U).$$

Suppose first $\beta < r$. On every instance with target in T , the mismatch query has expectation zero. Therefore $P_0 q \leq \kappa$: otherwise a prior mass at least r would be distinguished. Apart from a prior mass at most β , we then have $P_j q \leq 2\kappa$.

After L peels, the expected residual mass is at most $(1-r)^L \leq \eta/4$. Markov's inequality shows that at least half the prior puts residual mass at most $\eta/2$. Outside a union of exceptional sets of total mass at most $L\beta$, each of the L wrong-color masses is at most 2κ . The predictor assembled from the rectangle colors and an arbitrary residual label consequently has loss at most $2L\kappa + \eta/2 = \eta$ for prior mass at least $1/2 - L\beta$. If $\beta \geq r$, then $L\beta \geq L/\rho \geq \ln(4/\eta) > 1/2$, so the following inequality remains true trivially:

$$\max_g \Pr_{\pi}[g \text{ good}] + L \max_{q \in \mathcal{Q}} \Pr_{\pi}[q \text{ distinguishes}] \geq 1/2. \quad (16)$$

Apply the finite separation argument of Lemma 10 to the action pairs (g, q) , whose payoff on instance j is

$$\mathbf{1}_{g \text{ good for } j} + L \mathbf{1}_{q \text{ distinguishes } j}.$$

It gives laws $\mathcal{P}, \mathcal{Q}'$ satisfying

$$\Pr_{g \sim \mathcal{P}}[g \text{ good for } j] + L \Pr_{q \sim \mathcal{Q}'}[q \text{ distinguishes } j] \geq 1/2 \quad \text{for every } j.$$

On $\mathcal{B}(\mathcal{P})$, the second probability is greater than $1/(4L)$. This proves the required coverage.

The continuum of possible marginals creates no compactness gap in this minimax step. Both the predictor set and $\mathcal{Q}$ are finite. An instance is relevant only through the finite vector of Boolean good-solution and distinguishing-query indicators. There are finitely many such types. Choose one representative of each realized type and apply the finite argument to them. Its inequalities then hold for all original instances. No discretization in total variation and no factor polynomial in $|X|$ is introduced. $\square$

What the 2017 characterization actually says. In Feldman [6, Theorem 4.8], the randomized upper bound, in the notation used here, is

$$\text{RQC}(\mathcal{Z}, \text{STAT}(\kappa/3), \alpha - \delta) = O\left(d \frac{R_{\text{KL}}(\mathcal{D})}{\kappa^2} \log \frac{R_{\text{KL}}(\mathcal{D})}{\kappa\delta}\right), \quad d = \text{RSD}_{\kappa_1}(\mathcal{Z}, \kappa, \alpha). \quad (17)$$

The displayed asymptotic expression uses the usual positive-radius conventions; a ceiling and a maximum with one handle degenerate values. Here $R_{\text{KL}}(\mathcal{D}) = \inf_Q \sup_{P \in \mathcal{D}} \text{KL}(P\|Q)$. Taking uniform Q on Z proves $R_{\text{KL}} \leq \ln |Z|$: expand $\text{KL}(P\|Q) = \ln |Z| + \sum_z P(z) \ln P(z)$ and use $\ln P(z) \leq 0$. Thus the relevant domain term is $\ln(2|X|)$, not $|X|$ or the logarithm of a discretized instance count. The deterministic theorem has a different set-cover logarithm; it must not be substituted here.

For clarity, the mechanism behind (17) does not depend on a hidden finite-instance enumeration. Maintain a reference Q starting uniform on Z . A query separated from the true P by κ supplies a bounded direction with a signed expectation gap of order κ . For $f \in [-1, 1]$ and update $Q'(z) \propto Q(z)e^{\lambda f(z)}$, direct expansion gives

$$\text{KL}(P\|Q') - \text{KL}(P\|Q) = -\lambda \mathbb{E}_P f + \log \mathbb{E}_Q e^{\lambda f} \leq -\lambda(\mathbb{E}_P f - \mathbb{E}_Q f) + \lambda^2/2.$$

The final inequality follows by twice differentiating $\log \mathbb{E}_Q e^{\lambda f}$: its second derivative is a variance of a variable in $[-1, 1]$, at most one. Choosing λ proportional to κ decreases KL by $\Omega(\kappa^2)$. There are therefore $O(\ln |Z|/\kappa^2)$ updates. At each reference, sampling $O(d \ln(T/\delta))$ covering queries prevents an undetected bad instance over T updates, by $(1 - 1/d)^s \leq e^{-s/d}$ and a union bound. This proves the parameter dependence and extends the argument to all realizable marginals here.

At fairness $1/4$ and constant $\delta < 1/4$, this yields constant success probability. Fresh repetitions and a single loss query to verify each candidate amplify success with $O(\ln(1/\delta'))$ repetitions and loss slack $O(\kappa)$. This reasoning remains valid for history-dependent oracles by conditioning on earlier runs. The verifiable-search Theorem 4.11 in Feldman [6] is stated using a different statistical dimension over references admitting no good solution; its dimension must not be identified automatically with the fairness- $1/4$ quantity above. In either route verification does not erase the R_{KL}/κ^2 term of that generic simulation. The direct learner of Theorem 9 gives the sharper bound without it.

7 Cap mixtures and computation from an explicit table

The linear program in Lemma 10 has exponentially many possible rectangles. A fixed-dimensional template makes a different sampler available. Exact radial isotropy is used to state its clean geometric constant; the algorithmic corollary uses a rational approximation and a finite net. Exact real transformations and finite-bit computation are not identified.

Theorem 16 (Pointwise cap mixture and table-polynomial implementation). *Fix $d \geq 1$. Let F have a strict rank- d representation and no negative 2×2 , and let A be any monotone flip of its*

negative entries. Perturb the row vectors once into linear general position without changing F . At any nonempty survivor set, use the uniform row prior ν . There is a cap sampler, depending on the surviving rows but not on the unknown marginal, such that

$$\forall x \in U, \quad \mathbb{E}[\mathbf{1}_S(x)\nu(T)] \geq \frac{c_d}{8d}, \quad c_d = \Pr_{u \sim \text{Unif}(S^{d-1})}[u_1 \geq \sqrt{1 - 1/(2d)}]. \quad (18)$$

Here $c_1 = 1/2$. Thus Theorem 9 applies with $\rho = 8d/c_d$ in the ideal real-arithmetic model.

For each fixed d , given the explicit table of A and rational template vectors of bit length at most b , there are randomized and deterministic implementations, polynomial in $|X|, |H|, b, 1/\epsilon$, with

$$m = O_d(\log |H| + \log(1/\epsilon)), \quad \tau = \Omega_d(\epsilon/\log(1/\epsilon)).$$

Queries are intersections of stored cap tests, or such tests split by one row label or one point's row labels. This is polynomial time in the supplied table, not in $n = \log_2 |X|$ alone.

Proof. We first make the once-only perturbation precise. Number the rows $j = 1, \dots, K$. Replace a row vector v_j by

$$v_j(t) = v_j + t(1, j, \dots, j^{d-1}).$$

Choose a positive rational upper bound t_0 so that every dot product retains its strict sign for $0 < t < t_0$. The finitely many original rational dot products and the perturbation dot products give such a bound by direct comparison. For each d -subset of rows, its determinant is a polynomial in t of degree d whose leading coefficient is a nonzero Vandermonde determinant. At most $d \binom{K}{d}$ values are forbidden. Testing that many plus one rational grid values in $(0, t_0)$ finds a valid value, with polynomial bit length and polynomial work for fixed d . When $K < d$, perturb all subsets of size at most K to independence instead, or work directly in their span. The same perturbation works for every later survivor set.

Suppose there are $k > d$ surviving rows. Any proper j -dimensional subspace contains at most j of them, strictly less than kj/d . The radial-isotropy proof of Theorem 3 therefore applies; its convex formulation is also proved in Appendix B. Transform and normalize the rows to vectors $v_h \in S^{d-1}$ with

$$\mathbb{E}_\nu v_h v_h^\top = I_d/d.$$

Transform points contragrediently and normalize them to z_x . All template signs remain strict. If $k \leq d$, retain *all* rows and reduce to their common span: the map $x \mapsto (\langle v_h, u_x \rangle)_{h \in V}$ and the row vectors $e_h \in \mathbb{R}^k$ give the same strict signs and exact isotropy in dimension k . No target is discarded by a subspace restriction.

Write the effective dimension as $j \leq d$, put $\alpha^2 = 1/(2j)$ and $\beta = \sqrt{1 - \alpha^2}$, and sample u uniformly on S^{j-1} . Define

$$C_u = \{x : \langle u, z_x \rangle \geq \beta\}, \quad T_u^\pm = \{h : \pm \langle u, v_h \rangle > \alpha\}.$$

For $Z = \langle u, v_h \rangle^2 \in [0, 1]$, isotropy gives $\mathbb{E}_\nu Z = 1/j$. Since

$$\mathbb{E} Z \leq \alpha^2 + \Pr[Z > \alpha^2],$$

we have $\nu(T_u^+) + \nu(T_u^-) \geq 1/(2j)$. Choose the larger side, of mass at least $1/(4j)$.

The geometry is strict. If $s = \langle u, v \rangle > \alpha$ and $t = \langle u, z \rangle \geq \beta$, then

$$\langle z, v \rangle \geq ts - \sqrt{1 - t^2} \sqrt{1 - s^2} > 0,$$

because $s, t > 0$ and $s^2 + t^2 > 1$. Replacing v by $-v$ proves the negative-side assertion. Thus $C_u \times T_u^+$ is positive in F , and $C_u \times T_u^-$ is negative in F .

A positive rectangle survives in A . A negative rectangle has a singleton side. If its row side is $\{h\}$, partition $C_u \cap U$ according to $h(x)$ in A and choose either color with probability $1/2$. If its point side is $\{x\}$, partition the row side according to $h(x)$ and choose either color with probability $1/2$. Empty outcomes are allowed. The singleton-point case is essential and is not replaced by a singleton-row assumption. Conditional on $x \in C_u$, its expected weighted coverage after this split is at least half the chosen side's mass, hence at least $1/(8j)$. Rotational invariance gives probability c_j of cap membership. Therefore the pointwise bound is $c_j/(8j) \geq c_d/(8d)$. To verify the last inequality, couple the first coordinate of uniform spheres by normalizing an increasing list of independent standard Gaussians. Its absolute value decreases as dimension grows, while $\sqrt{1 - 1/(2j)}$ increases. Symmetry supplies the factor $1/2$, including the case $j = 1$.

For finite-bit computation, Appendix B gives a deterministic rational transform with normalized covariance at least $3I_j/(4j)$. The preceding second-moment argument then gives total side mass at least $1/(4j)$, selected-side mass at least $1/(8j)$, and split coverage at least $1/(16j)$ whenever a point belongs to the cap. There is no claim that an exact irrational optimum is computed.

Use the finite direction set

$$W_j = \left\{ \frac{g}{\|g\|} : 0 \neq g \in (16j)^{-1}\mathbb{Z}^j \cap [-1, 1]^j \right\}, \quad J_j = |W_j| \leq (32j + 1)^j.$$

Repetitions of a direction cause no difficulty. Rounding a unit vector coordinatewise produces g with $\|g - z\| \leq 1/(32\sqrt{j})$; normalization changes it by at most twice this amount. Thus $\langle z, g/\|g\| \rangle \geq 1 - 1/(512j) > \beta$, where the last inequality follows from $\beta \leq 1 - 1/(4j)$. Every point is in at least one cap. Uniformly sampling W_j and the split color yields coverage at least $1/(16jJ_j)$. The common rational lower bound

$$r_d = \frac{1}{16d(32d + 1)^d} \tag{19}$$

works in all effective dimensions. All membership comparisons are rational: for example $\langle z, g/\|g\| \rangle \geq \beta$ means that the unnormalized dot product is positive and

$$\langle z_{\text{raw}}, g \rangle^2 \geq (1 - 1/(2j))\|z_{\text{raw}}\|^2\|g\|^2.$$

The row-side tests use the analogous strict squared comparison. Consequently no exact sampling of a continuous direction is required. Theorem 9 with r_d proves the randomized finite-bit bound.

For determinism, enumerate the at most $2J_j$ outcomes and query each $D(S)$, using the same tolerance and stopping check as Theorem 9 with r_d . Choose the outcome maximizing $b\hat{s}$, where $b = \nu(T)$. Coverage implies that some outcome has $bs \geq r_d u$. The chosen one has

$$bs \geq r_d u - 2\tau, \quad ab \geq r_d - 2\tau/u \geq 3r_d/4.$$

Thus both a and b are at least $3r_d/4$. Query its mismatch mass and peel or eliminate exactly as before. The target survives, and all error bounds remain valid. Each stage has progress at least $3r_d/4$, whereas the deterministic total-progress bound is B . There cannot be more than $\lceil 4B/(3r_d) \rceil + 1$ nonstopping stages. There are $O_d(1)$ queries per stage. This proves the deterministic claim, without fixing a successful private seed after seeing D . $\square$

For $d = 6$, $c_6 \approx 0.0003509774$ and $8d/c_d \approx 136760.93$. These numerical values describe the ideal sampler. The rational-net construction has worse, dimension-dependent constants but the same stated asymptotics. A low-rank template alone is insufficient: the no-negative- 2×2 condition is used exactly at the splitting step.

8 Proper learning without a dual-plane search

Every grid family contains the all-positive row: take intercept $b = 2N^2$. Proper learning therefore does not require testing random lines when the total negative mass is small. The first substantial negative rectangle supplies the missing localization information.

Theorem 17 (Proper RECTIFY). *Suppose H contains the all-positive hypothesis and has no negative 2×2 . Given a rectangle sampler of coverage r , it has a distribution-independent proper SQ learner of expected error at most ϵ , using*

$$m_{\text{proper}} \leq m_{\text{RECTIFY}}(\epsilon/4) + \lceil \log_2 |X| \rceil + \lceil \log_2 K \rceil + \lceil 8/\epsilon \rceil + 4 \quad (20)$$

queries at the tolerance of RECTIFY with accuracy $\epsilon/4$. For the incidence flips,

$$m_{\text{proper}} = O(\log M + \epsilon^{-1} + \log(1/\epsilon)), \quad \tau = \Omega(\epsilon/\log(1/\epsilon)).$$

The cap implementation is polynomial-time in the explicit table. Its finite-net deterministic variant is also proper and has the same asymptotic query bound, with constants depending on the template dimension.

Proof. Set $\theta = \epsilon/4$ and run RECTIFY at accuracy θ , stopping at its first proposed negative peel. Until then all assigned labels, and its fallback, are positive. If it stops normally without a negative peel, output the all-positive member of H . The error is at most $\theta/2$. Its budget-failure probability is at most $\theta/2$.

Suppose a negative rectangle $S \times T$ reaches a peeling answer. The acceptance calculation in Theorem 9 and $u > \theta/8$ give

$$D(S) > 5r\theta/128.$$

For $P_\theta = 1 + \lceil 4r^{-1} \ln(8/\theta) \rceil$ and $\tau = \theta/(24P_\theta)$,

$$\frac{D(S)}{\tau} > \frac{15}{16}rP_\theta > \frac{15}{4} \ln(8/\theta) > 18,$$

since $\theta < 1/16$. A peeling answer implies $D(S \cap \{h^* = +1\}) \leq 3\tau$. Hence the negative mass in S exceeds 15τ .

If $S = \{p\}$, then $h^*(p) = -1$ and p is known. Otherwise the negative rectangle has a single row h_0 . Query its full loss, accepting it when the answer is at most $\epsilon/2$. An accepted candidate has true loss at most $\epsilon/2 + \tau$. If it is rejected, $h_0 \neq h^*$, since the target's answer would be at most τ . The two rows share at most one negative point. Therefore the negative measure in S is concentrated at a unique point p and has mass greater than 15τ . Locate p by binary search over an arbitrary ordering of S : for a half $B \subseteq S$, query $D(B \cap \{h^* = -1\})$. The answer is at most τ when $p \notin B$, and greater than 14τ when $p \in B$. Threshold 4τ safely determines the half. This takes at most $\lceil \log_2 |X| \rceil$ queries.

Now let $V_p = \{h : h(p) = -1\}$. Its negative supports away from p are pairwise disjoint; otherwise two rows share two negative points. Query the target's negative mass away from p ,

$$\beta = D(\{x \neq p : h^*(x) = -1\}).$$

If the answer exceeds $\epsilon/4$, then $\beta > \epsilon/4 - \tau$. For any half $W \subseteq V_p$, define

$$Q_W(x, y) = \mathbf{1}_{\{y = -1\}} \mathbf{1}_{\{x \neq p\}} \mathbf{1}_{\{x \in \bigcup_{h \in W} \{h = -1\}\}}.$$

Its expectation is exactly β if $h^* \in W$ and exactly zero otherwise. Thus threshold $\epsilon/8$ distinguishes the two cases, since $\tau < \epsilon/16$. Binary search finds the target row in at most $\lceil \log_2 K \rceil$ queries. If two indexed rows coincide and contain only p as a negative point, this branch cannot occur; in the positive-tail branch the target row is unique.

If the answer for β is at most $\epsilon/4$, then $\beta \leq \epsilon/4 + \tau$. Test up to

$$J = \lceil 8/\epsilon \rceil + 1$$

distinct members of V_p , again accepting an estimated full loss at most $\epsilon/2$. If fewer than J members exist, these tests include the target. Otherwise their negative masses outside p sum to at most one, by disjointness. One tested row h has such mass at most $1/J < \epsilon/8$. It agrees at p , and therefore

$$L_{D,h^*}(h) \leq \beta + D(\{x \neq p : h(x) = -1\}) < 3\epsilon/8 + \tau.$$

Its answer is less than $\epsilon/2$, because $2\tau < \epsilon/8$. Some test consequently passes. Every accepted row has loss at most $\epsilon/2 + \tau$.

All nonfailure outcomes are proper and have loss at most $\epsilon/2 + \tau$. Any budget failure can output the all-positive row, with loss at most one. Expected loss is bounded by

$$\epsilon/2 + \tau + \theta/2 < \epsilon.$$

The extra query counts give (20). Using the deterministic net implementation removes the budget-failure event and leaves the same proper completion. Its computation consists of cap tests, row-label tests, and unions of explicitly listed row supports, all polynomial in the table size. $\square$

The all-light case requires no aspect-ratio argument. At the first negative peel, either the singleton candidate is already good, or its intersection with the target contains one heavy negative atom. The other members of the star at that atom have disjoint negative supports. This is a proper-learning theorem, not an exponential proper-improper separation.

9 Rectangle geometry: tight examples and non-necessity

Theorem 18 (Global projective-plane rectangle ratio). *For the point–line incidence class of $\text{PG}(2, q)$, over any prime-power order $q \geq 2$ for which the plane is defined,*

$$\frac{1}{8(q+1)} \leq \text{rect}(H_q) \leq \frac{q}{(q+1)^2}.$$

Thus the infimum over all product distributions is $\Theta(1/q)$, not merely the maximum rectangle under the uniform product measure.

Proof. We prove the lower bound for any bipartite incidence graph whose degrees on both sides are at most $\Delta \geq 1$. Fix arbitrary product weights $\mu \times \nu$ and let p be their total incidence mass. If $p \geq 1/2$, some point has incident row mass at least $1/2$. At most Δ rows meet it, so one row has prior mass at least $1/(2\Delta)$. One of that row's two color classes has point mass at least $1/2$. This gives a monochromatic rectangle of mass at least $1/(4\Delta)$.

If $p < 1/2$, select each point independently with probability $t = 1/(2\Delta)$, obtaining S , and let T be all rows with no neighbor in S . The rectangle $S \times T$ is nonincident. For a nonincident pair (x, h) ,

$$\Pr[x \in S, h \in T] = t(1-t)^{\deg h} \geq t(1-\Delta t) = 1/(4\Delta).$$

The inequality $(1 - t)^j \geq 1 - jt$ follows by induction. Averaging the rectangle mass gives at least $(1 - p)/(4\Delta) > 1/(8\Delta)$. Hence some rectangle has that mass. Taking $\Delta = q + 1$ proves the universal lower bound.

For the upper bound choose uniform weights. Let $K = q^2 + q + 1$, $k = q + 1$, and let P be the incidence matrix. The identities $P^\top P = qI + \mathbf{1}\mathbf{1}^\top$ and $P\mathbf{1} = k\mathbf{1}$ give

$$C = P - (k/K)\mathbf{1}\mathbf{1}^\top, \quad \|C\|_{\text{op}} = \sqrt{q}.$$

A nonincident rectangle of sizes a, b satisfies $(k/K)ab \leq \sqrt{q}\sqrt{ab}$, so $ab/K^2 \leq q/k^2$. An incident rectangle has a singleton side and area at most k . Its mass is at most $1/K \leq q/k^2$, since $qK - k^2 = q^3 - q - 1 > 0$. This establishes the claimed upper bound on the infimum. $\square$

What is and is not tight. Theorem 2 and the balanced target-dependent marginals of Theorem 4 give $m/\tau^2 = \Omega(q) = \Omega(\rho)$ for this family. At a fixed positive tolerance this yields $m = \Omega(\rho)$. Theorem 9, however, uses $\tau = \Theta_\epsilon(1/\rho)$ and has $m/\tau^2 = O_\epsilon(\rho^3 \log K)$. Thus the geometry is sharp and a linear inverse-rectangle dependence is unavoidable at fixed tolerance, but the joint (m, τ) dependence of Theorem 9 is *not* established as tight up to logarithms. Lowering tolerance cannot be ignored in a tightness claim.

Theorem 19 (Small rectangle ratio does not prevent SQ learning). *For odd n , let*

$$X_n = \{-1, +1\}^n, \quad H_n = \{x \mapsto \text{sign}\langle w, x \rangle : w \in \{-1, +1\}^n\}.$$

Then $\text{dc}(H_n) \leq n$, $\text{rect}(H_n) \leq e^{-n/4}$, and there is one distribution-independent SQ learner with $m = O(n^3/\epsilon^2)$ and $\tau = \epsilon/(8n)$.

Proof. All integer inner products are odd and nonzero, so the identity embedding proves the dimension claim. Distinct parameter vectors give distinct rows: conditioning on all but coordinate i shows

$$\mathbb{E}_x[x_i \text{sign}\langle w, x \rangle] = w_i \Pr[Z_1 + \dots + Z_{n-1} = 0],$$

where the Z_j are independent fair signs and the probability is positive. The conditional difference of the two signs is two exactly when that even sum is zero, and is zero otherwise. Thus uniform parameters are also a uniform prior on the distinct hypotheses. We first bound every rectangle under this uniform product measure. A negative rectangle gives two subsets of the cube with all cross Hamming distances at least $t = (n + 1)/2$. For a positive rectangle, negate one of its sides to get the same separation condition. Negation preserves cardinality.

Let A, B be nonempty subsets separated in this way. For uniform Z on the cube, set $f(Z) = \text{dist}_{\text{Ham}}(Z, A)$ and $s = \mathbb{E}f$. Changing one coordinate changes f by at most one. The elementary bounded-differences calculation proved in Appendix C gives

$$\Pr[f - s \leq -a] \leq e^{-2a^2/n}, \quad \Pr[f - s \geq a] \leq e^{-2a^2/n} \quad (a \geq 0).$$

Writing $\alpha = |A|/2^n$, $\beta = |B|/2^n$, if $0 \leq s \leq t$ then

$$\alpha\beta \leq \exp\{-2(s^2 + (t - s)^2)/n\} \leq e^{-t^2/n} \leq e^{-n/4}.$$

If $s \geq t$, already $\alpha \leq e^{-2t^2/n}$ and $\beta \leq 1$. Empty rectangles have zero mass. Taking the maximum under this one product measure proves the rectangle upper bound.

For learning, set $z = x/\sqrt{n}$, $\gamma = 1/n$, and optimize

$$F(v) = \mathbb{E}_D(\gamma - y\langle v, z \rangle)_+ \quad \text{over } \|v\| \leq 1.$$

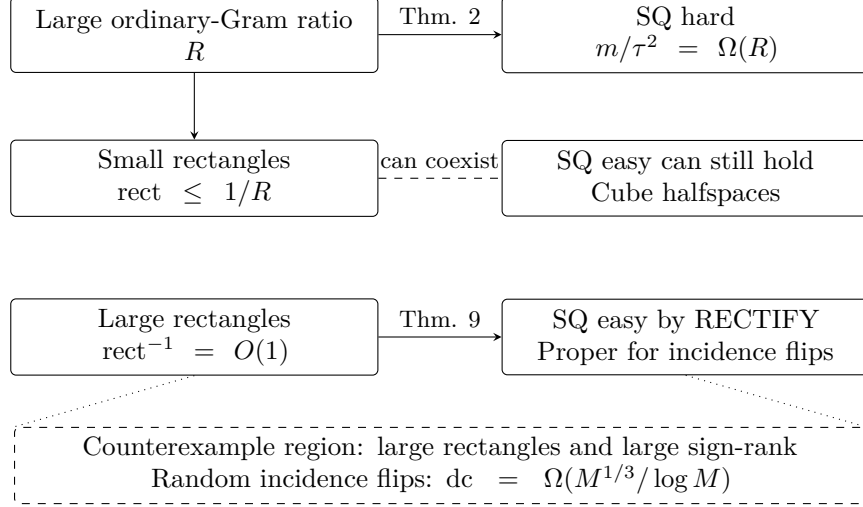


Figure 2: The valid implications and compatible examples. There is no arrow from small rectangles to SQ hardness. A fixed-marginal spectral certificate implies both conclusions separately. The dotted connections locate an example, rather than claim that large rectangles force large sign-rank.

The target $v^* = w^*/\sqrt{n}$ has $F(v^*) = 0$. At a current v , a valid subgradient is

$$g(v) = \mathbb{E}[-yz \mathbf{1}_{\{y\langle v, z \rangle < \gamma\}}], \quad \|g(v)\| \leq 1.$$

Query each coordinate using the bounded function $q_i(x, y) = -yx_i \mathbf{1}_{\{y\langle v, x/\sqrt{n} \rangle < \gamma\}}$ and divide its answer by $\sqrt{n}$. The resulting vector is $g(v) + e$ with $\|e\| \leq \tau$, regardless of the coordinatewise answer choices.

Start $v_0 = 0$, take $T = \lceil 100/(\epsilon^2 \gamma^2) \rceil$, step $a = T^{-1/2}$, and update by projection of $v_t - a(g_t + e_t)$ onto the unit ball. For any point v^* in that ball, projection cannot increase distance to v^* . Expanding the squared distance and using convexity yields

$$\frac{1}{T} \sum_{t \leq T} F(v_t) \leq \frac{1}{2aT} + \frac{a}{2}(1 + \tau)^2 + 2\tau \leq \frac{5}{2\sqrt{T}} + 2\tau \leq \epsilon\gamma/2.$$

Here $\|v_t - v^*\| \leq 2$, $\tau = \epsilon\gamma/8 < 1$, and $\sqrt{T} \geq 10/(\epsilon\gamma)$. Jensen's inequality gives the same hinge bound for $\bar{v} = T^{-1} \sum_t v_t$. Every classification mistake of $\text{sign}\langle \bar{v}, z \rangle$ incurs hinge loss at least γ , including a zero score when the true label is negative. Thus its error is at most $\epsilon/2$. There are nT scalar queries. This proves a deterministic, adversarial-oracle guarantee stronger than the stated expected-error requirement. $\square$

For contrast, all parities on $\{-1, +1\}^n$ have an orthonormal label Gram matrix under the uniform marginal: the product of distinct characters is a nonconstant character, whose mean cancels by flipping one coordinate. Theorem 2 therefore gives $m/\tau^2 = \Omega(2^n)$ at any fixed accuracy below $1/4$. The same Walsh orthogonality bounds every uniform monochromatic rectangle mass by 2^{-n} . Small rectangles can therefore occur on either the easy or the hard side of SQ learning.

Why the spectral certificates miss the counterexample. For a fixed-marginal label-Gram certificate G on K hypotheses, a monochromatic rectangle $S \times T$ satisfies

$$\|G\|_{\text{op}} \geq \mathbb{E}_\mu \left(\sum_{h \in T} h(x) / \sqrt{|T|} \right)^2 \geq \mu(S)|T|.$$

Hence every such rectangle under $\mu \times \text{Unif}(H)$ has mass at most $\|G\|_{\text{op}}/K$. A large ordinary-Gram ratio therefore implies small rectangles *and*, separately, SQ hardness. The arrow from small rectangles to hardness is false by Theorem 19. The balanced projective-plane certificate is a joint-distribution certificate, not one of these bounded ordinary-Gram ratios. For the flip family, Theorems 2 and 9 give $\mathfrak{J}(H_A) = O_\epsilon(\log M)$ while $\text{dc}(H_A) = \Omega(N/\log M)$. This is the promised failure of joint-certificate completeness.

Theorem 20 (Marginal adaptation, with its query-budget qualification). *Any deterministic learner of a high-dimension flip class obeys*

$$u \geq \frac{\ln \text{dc}(H_A) - \ln(m+1)}{\ln(\lceil 2/\tau \rceil + 1)},$$

where u is the marginal-branching depth defined in Theorem 5. In particular, for $m \leq (\log M)^C$ and all sufficiently large N ,

$$u = \Omega_C\left(\frac{\log N}{\log(2 + 1/\tau)}\right).$$

At fixed positive tolerance, the $\Omega(\log N)$ requirement matches the $O_d(\log M)$ marginal-query upper bound of the deterministic finite-net learner, at fixed accuracy.

Proof. Take logarithms of Theorem 5, $\text{dc}(H_A) \leq (m+1)(\lceil 2/\tau \rceil + 1)^u$. The high-dimension event gives $\ln \text{dc}(H_A) \geq \ln N - \ln \log M + O(1)$. Under the stated query budget the numerator is at least $\frac{1}{2} \ln N$ for large N . The upper bound is Theorem 16. All constants depend only on the displayed fixed parameters. $\square$

The query-budget qualification cannot be dropped: querying the correlation with every row is a deterministic correlational learner with $u = 0$ and $m = K$. Randomization is not intrinsically necessary for the separation, because the finite-net enumeration is deterministic. What is necessary for a small-query deterministic learner here is adaptation to marginal information. At inverse-polylogarithmic tolerance the proved lower bound loses a $\log \log M$ factor; no sharper matching statement is made.

10 Possible objections

Is this Feldman’s 2017 separation? No. Feldman’s finite-field line class is easy for every separately fixed, known marginal but is hard for one distribution-independent SQ learner [6, Theorems 7.7–7.8]. RECTIFY is one algorithm for every marginal of a different, real-grid incidence-flip family. The unknown marginal is learned through valid SQ answers, not supplied as side information. Theorem 4 independently verifies the earlier type of separation for projective planes.

Does knowing A constitute a loophole? The learner knows the concept class, as permitted by the statement being refuted. Its table is not charged to query count. This is an important resource asymmetry, not access to the unknown target or distribution. Section 11 shows why merely charging source-code length does not repair it.

Is the class natural or efficiently explicit? The high-sign-rank selection is obtained by counting, not by an efficiently computable formula for its entries. The short exhaustive generator below is computable but not an efficient explicit construction. The computational explicitness problem remains related to Problem 4.1 of Hatami et al. [2].

Is the learner efficient? For fixed template dimension, Theorem 16 gives polynomial time in the supplied truth table and its rational representation; Theorem 17 also makes the output proper. The table itself has exponentially many bits in the example encoding length. No running-time polynomial in n alone is claimed.

Does randomization matter? The general rectangle theorem uses a randomized mixture. For these fixed-dimensional templates a finite net can be enumerated, giving a deterministic learner with the same fixed-dimension query asymptotics. Theorem 20 establishes a marginal-adaptation requirement under a small-query budget, not an unconditional randomization requirement.

11 What survives for SGD and resource-bounded learners

The SQ theorem does not settle the benign-SGD question. A generic SQ algorithm can store or regenerate its class description for free before issuing any query. Ordinary gradient dynamics have a fixed network, initialization, and objective. The SQ-to-differentiable-learning simulation of Abbe et al. [8] uses engineered models and initialization; its computational network size depends on the simulated computation, not solely on the number of SQ calls. Its ReLU alternative also allows fixed weights. Our table-polynomial implementation remains exponential-time in n when the table is supplied explicitly, and therefore does not establish a small plain network by that simulation.

For reference, a robust population-gradient method with S parameters, T calls, per-example coordinate bound B_g , and adversarial Euclidean error tolerance δ is simulated by

$$m = TS, \quad \tau = \delta/(B_g\sqrt{S}).$$

Query each gradient coordinate divided by B_g , then multiply back: the coordinate errors are at most $\delta/\sqrt{S}$, so their squared sum is at most δ^2 . Induction over updates proves the simulation. Conversely the adaptive objective $J_q(t; x, y) = tq(x, y)$ has population derivative $\mathbb{E}q$, realizing a scalar SQ. This converse permits changing the objective; it is not the fixed logistic-loss process of the SGD question.

The remaining SGD question is exactly the one with $X = \{-1, +1\}^n$, a fully connected bias-free ReLU network, independent layer- i Gaussian initialization of variance $1/n_{i-1}$, online gradients of $\log(1 + e^{-yf_\theta(x)})$, one fixed step size, and classification by the specified tail sum. Does a common expected-error guarantee at $\epsilon < 1/4$ imply $\text{dc}(H) \leq CTS$ or $\text{dc}(H) \leq \text{poly}(T, S)$? A sampled gradient is not a worst-case approximate population gradient, and success under its sampling law does not imply robustness to every vector perturbation. No answer for those particular dynamics is established here.

Proposition 21 (Representation does not give the required dynamics). *Pad the grid domain to a cube of dimension $n = \lceil \log_2(2N^3) \rceil$, giving every padding point label +1. Every member is represented exactly by a bias-free ReLU network with widths $(n, 2n, N + 1, 1)$ and at most*

$$S = 2n^2 + 2n(N + 1) + (N + 1)$$

weights. Its common dimension complexity is at most $2N + 1$. This proposition is an upper bound, not a claim that S parameters are necessary.

Proof. The first hidden layer computes $\sigma(x_i), \sigma(-x_i)$ for every coordinate. These give x_i by subtraction and the constant $c = \sigma(x_1) + \sigma(-x_1) = 1$ on the cube. For every negative point p , a second-layer unit computes

$$\sigma\left(\sum_i p_i x_i - (n - 1)c\right).$$

At $x = p$ its value is one; at every other cube point the sum is at most $n - 2$ and the unit is zero. Pass c through one additional unit and output c minus twice the sum of the negative-point units. Pad unused units by zero weights. There are at most N negative points, and counting all fully connected weights gives the claimed S . The moment-curve proof in Theorem 11 still applies to the padded domain. Padding preserves every dimension lower bound by restriction; it preserves a rectangle lower bound up to a factor two, because either padding carries at least half the point mass or the original domain does. $\square$

A typical flip table has $I = \Theta(N^4)$ unconstrained bits. This does not by itself prove a lower bound of N on the number of unrestricted real network parameters: a real parameter has no fixed bit budget. Under a separate b -bit-per-weight convention, a fixed S -weight architecture has at most 2^{bS} parameter choices, so representing all 2^N independently specified patterns on a fixed line would require $bS \geq N$. That bounded-precision counting statement concerns an entire family of patterns, not a necessary parameter count for every fixed row of one fixed A . It is not used to answer the SGD question.

Theorem 22 (Source-description length alone does not repair the implication). *Interpret ℓ as the bit length of a finite learner program, including literal hard-coded constants and data, but do not charge running time or generated workspace. There is a computable sequence of incidence-flip classes and learners with*

$$n, \ell, m = O(\log N), \quad 1/\tau = O_\epsilon(1), \quad \text{dc}(H) = \Omega(N/\log N).$$

Consequently no universal bound $\text{dc}(H) \leq \text{poly}(m, 1/\tau, n, \ell)$ holds in this interpretation.

Proof. Let $K = 2N^3$, $b_N = \lceil \log_2(4K) \rceil$, and $d_N = \lfloor N/(128b_N) \rfloor$. For sufficiently large N , the counting bound in Theorem 11 guarantees a flip matrix of sign-rank greater than d_N : indeed $\log_2(4eK) \leq 2b_N$, so

$$2Kd_N \log_2(4eK) \leq N^4/16 < I.$$

Select the lexicographically first such matrix. This selection is computable. For a proposed matrix A , the predicate $\text{signrank}(A) \leq d_N$ is the truth of the finite real sentence

$$\exists U \in \mathbb{R}^{K \times d_N}, V \in \mathbb{R}^{d_N \times K} \quad \bigwedge_{i,j} A_{ij} \sum_{s=1}^{d_N} U_{is} V_{sj} > 0.$$

Decision of finite polynomial real sentences is the classical real-closed-field decision theorem. We use this explicit imported decidability result, not a claimed practical low-rank solver; a formal constructive treatment is given by Mahboubi and Cohen [13]. Enumerate the finitely many flip tables and decide the displayed sentence for each until the first failure. The counting argument ensures termination. Small N can be handled by any fixed table.

A single fixed program performs this generation and then runs the finite rational rectangle-mixture algorithm. Encode N in $O(\log N)$ bits. Fix an accuracy such as $1/16$, so that all accuracy constants have constant-length descriptions. Conservative rational ceilings can replace the logarithmic constants in Theorem 9, making every step an ordinary finite computation. The table generation happens before the first SQ; it has no query cost. The learner therefore has $m = O(\log K)$ and constant positive tolerance, while the chosen class has $\text{dc} > d_N = \Omega(N/\log N)$. Its program description has size $O(\log N)$. The generated table occupies far more space, which this interpretation does not charge. The displayed polynomial implication is contradicted. $\square$

This does not produce an efficient explicit counterexample. It instead shows that a resource-aware repair must charge computation, generated memory, or an expanded circuit description, not just the length of the program that generates them. One genuinely stronger question is whether charging total preprocessing and query-evaluation time t yields a bound polynomial in $(m, 1/\tau, n, \ell, t)$ in a specified finite-bit model. We state this as an open question, not as an established or uniquely natural repair.

Proposition 23 (What the classical-method barrier actually bounds). *For the orientation in which rows are hypotheses, let*

$$m_{\text{avg}}(A) = \inf_{\mu \in \Delta(X)} \sup_{\text{strict unit-vector representations of } A} \min_h \mathbb{E}_{x \sim \mu} |\langle v_h, u_x \rangle|.$$

Then

$$\sqrt{\text{VC}(H)} \leq m_{\text{avg}}(A)^{-1} \leq \text{rect}(A)^{-1}.$$

In particular, if $\text{rect}(A)^{-1} \leq \text{poly}(n)$, these three classical lower-bound parameters cannot certify superpolynomial sign-rank. This does not upper-bound sign-rank itself and does not imply that every SQ-easy class has large rectangles.

Proof. Fix a point prior μ . Apply the finite separation argument of Lemma 10 to the transposed matrix, obtaining a distribution λ over monochromatic rectangles $S_R \times T_R$ with

$$\forall h, \quad \mathbb{E}_R[\mathbf{1}_{T_R}(h)\mu(S_R)] \geq r, \quad r = \text{rect}(A).$$

In the Euclidean space indexed by rectangles, put

$$a_h(R) = c_R \sqrt{\lambda_R} \mathbf{1}_{T_R}(h), \quad b_x(R) = \sqrt{\lambda_R} \mathbf{1}_{S_R}(x).$$

Their norms are at most one, and their inner product is A_{hx} times the probability that R covers (h, x) . These products may be zero, so this alone is not a strict representation. Add an orthogonal strict unit-vector representation scaled by $\sqrt{\delta}$, and scale the displayed vectors by $\sqrt{1 - \delta}$. Every resulting product is strict with the required sign, and its average absolute value in each row is at least $(1 - \delta)r$. Pad row norms and point norms separately in orthogonal private coordinates to make them one without changing any cross inner products. Letting $\delta \downarrow 0$ proves $m_{\text{avg}}(A) \geq r$.

For the VC bound, suppose v points are shattered and use their uniform point prior. In any strict unit-vector representation, select one row h_σ for each sign pattern $\sigma \in \{-1, +1\}^v$. If the minimum row-average margin is a , then

$$va \leq \sum_{i=1}^v \sigma_i \langle v_{h_\sigma}, u_i \rangle \leq \left\| \sum_{i=1}^v \sigma_i u_i \right\|.$$

Averaging the square of the last norm over independent fair signs gives v , because cross terms cancel. Some sign pattern has norm at most $\sqrt{v}$, so $a \leq 1/\sqrt{v}$. Take the supremum over representations and the infimum over priors. For $v = 0$ the inequality is immediate. This proves the chain, including the strict-sign correction. $\square$

The chain and its original average-margin interpretation are due to Hatami et al. [2]; the proof is included to fix the orientation and strict-sign issue. A counterexample obtained with a polynomial inverse rectangle bound cannot be certified as high-sign-rank by those numerical parameters. Counting is already a way around that barrier, as Theorem 11 demonstrates. Therefore the barrier does not rescue the program-length conjecture. For an *efficiently explicit* sequence

it motivates seeking a method beyond those parameters, in the direction of Hatami et al. [2, Problem 4.1].

The recent topological methods provide a distinct parameter, not one proved bounded by this chain. The checked results do not supply the missing efficient total construction: Frick et al. [11] prove near- $2k$ bounds for certain *partial* Gap Hamming matrices, while G6ös et al. [10] bound the sign-rank of total k -Hamming-distance matrices by $2^{O(k)}$ independently of the bit dimension. These facts neither rule out a future topological route nor prove that it evades the rectangle obstruction on the particular flip family. No such explicit corollary is claimed here.

12 Open problems and limits

The statistical-query implications to dc , $\text{dc}_{1/2}$ and $\text{dc}^{C\epsilon}$ are refuted. The proper-improper issue for this family is settled on the proper-learning side. Neither result settles learning by the particular online Gaussian-initialized bias-free ReLU dynamics.

An efficiently computable, total, high-sign-rank matrix with polynomial inverse rectangle ratio remains the explicit-construction problem investigated here. The exhaustive generator in Theorem 22 is not a solution under that computational meaning of explicitness. A parameter measuring rectangle-type product-distribution structure may still characterize distribution-independent SQ complexity, but rect^{-1} alone cannot: cube halfspaces and parities both have exponentially small rectangles, while only the latter have the corresponding SQ obstruction. Such a characterization would need an additional feature or an optimized representation of the task.

The joint dependence on query count and tolerance in Theorem 9 remains unoptimized. The projective-plane example proves a linear lower bound on m/τ^2 , not a matching lower bound for the theorem’s full parameter tradeoff. Finite-bit computation or expanded-circuit size, rather than source-program length alone, is necessary for a meaningful resource-aware repair. No general polynomial-dimension theorem under those stronger charges is proved. Finally, the Lean sources have not been compiled in this environment; they are not a verified formalization of the new main results.

A Independent referee derivation and endpoint audit

We rederived Theorem 9 without using a favorable oracle. The result agrees with its earlier constants. In the table below $u = D(U)$, $s = D(S)$, $a = s/u$, $b = \nu(T)$, $w = D(S \cap \{h^* \neq c\})$, and hats denote oracle answers. “Pathwise” means that no random-oracle assumption is used. The conditioning for mixture expectations includes the returned $\hat{u}$ and all earlier history, but precedes the fresh rectangle draw.

	Inequality or invariant	Adverse valid answer or justification	Randomness used
R1	$\tau \leq \epsilon/8, \tau \leq r\epsilon/64$	Direct substitution of P ; no reply.	None
R2	Nonstop $u > \epsilon/4 - \tau \geq \epsilon/8$	$\hat{u} = u + \tau$ can keep the smallest u active.	Pathwise
R3	$\tau/u \leq r/8$	R1–R2.	None
R4	Acceptance gives $s \geq ru/2 - (1 + r/2)\tau$	$\hat{s} = s + \tau, \hat{u} = u - \tau$ favor accepting a small set.	Pathwise
R5	Accepted $a \geq 5r/16 \geq r/4$	Substitute R3 into R4.	None

	Inequality or invariant	Adverse valid answer or justification	Randomness used
R6	$a \geq 3r/4$ gives $\hat{s} - r\hat{u}/2 \geq ru/16 > 0$	$\hat{s} = s - \tau$, $\hat{u} = u + \tau$ oppose acceptance.	Pathwise
R7	Target in T implies $w = 0$, $\hat{w} \leq \tau < 2\tau$	Maximal reply $\hat{w} = w + \tau$.	Pathwise
R8	$\sum_{\text{elim}} b \leq \sum_{\text{elim}} -\ln(1-b) \leq \ln K$	Target retained; normalized target weight grows from $1/K$ to at most one.	Pathwise
R9	Before peel j : $\epsilon/8 \leq (1-r/4)^{j-1} \leq e^{-(j-1)r/4}$	R2, R5, and $\ln(1-z) \leq -z$.	Pathwise
R10	Number of peels at most P	Take logarithms in R9.	None
R11	A peel has $w \leq 3\tau$; total wrong mass $\leq 3P\tau = \epsilon/8$	$\hat{w} = w - \tau$ permits the largest wrong mass. Peeled sets are disjoint.	Pathwise
R12	Normal stop: $u \leq \epsilon/4 + \tau$; loss $\leq \epsilon/2$	$\hat{u} = u - \tau$ permits the largest residual mass. Empty residual has mass zero.	Pathwise
R13	On acceptance, $G = a$ or b , so $G \geq ab$	Both admissible actions obey this, irrespective of the mismatch reply.	Pathwise
R14	$\mathbb{E}[ab \mid \text{past}] \geq r$	Integrate pointwise mixture coverage against D conditioned on U .	Fresh learner draw
R15	$\mathbb{E}G \geq r - (3r/4)\mathbb{E}b \geq r/4$	Rejection is confined to $a < 3r/4$ by R6; $b \leq 1$.	Fresh learner draw
R16	$\sum G \leq \ln K + \ln(8/\epsilon) + 1 = B$ before budget termination	R8; peel ratios telescope excluding the last, whose $a \leq 1$. No logarithm of zero.	Pathwise
R17	$e^{-g} \leq 1 - (1 - e^{-1})g$ for $0 \leq g \leq 1$	Convexity chord. After normal stopping put $G = 1$.	None
R18	$\mathbb{E}[e^{-G} \mid \text{past}] \leq e^{-r/8}$; $\mathbb{E}e^{-\sum G} \leq e^{-rR/8}$	R15, R17 and conditional expectation induction.	Learner draws; oracle arbitrary
R19	$\Pr[\text{budget failure}] \leq e^{B-rR/8} \leq \epsilon/2$	R16 and Markov's inequality.	Learner draws; oracle arbitrary
R20	$\mathbb{E}L \leq \epsilon/2 + \Pr[\text{budget failure}] \leq \epsilon$	R12 and $0 \leq L \leq 1$.	Learner draws; oracle arbitrary

A randomized oracle can be included by conditioning on its past private coins; each displayed response bound still holds for every permitted answer. No step averages away an adversarial error. The drift proof allows the oracle to choose either permissible action after seeing the current query. It does not give an oracle access to future learner randomness.

Counting audit. There are exactly $N(N+1)/2$ empty template rows: row (a, b) has no incidence precisely when $b > 2N^2 - a$. They are all-positive rows of every flip. Their weights k_h/I in Theorem 12 are zero; no uniform distribution on an empty set is formed. For nonempty rows these weights sum to one. For a fixed embedding, loss minimization ranges over finitely many attained sign patterns, so the infimum is a minimum. The weighted incidence mistake count is an integer divided by I ; some embedding has value at most its expectation bound. The same embedding is used for every row, with row-specific weights, exactly as required by the definition.

The working convention is $\text{sign}(0) = +1$. A positive row-specific bias in one common added coordinate turns zero scores positive without changing any negative score. Thus the strict rank

in the counting argument is $d + 1$, and the event is $\text{dc}^\eta < s_\eta$, not $\text{dc}^\eta \leq s_\eta$. If the sign convention is $\text{sign}(0) = -1$, use a negative bias. If zero predictions are counted as errors, replacing them by $+1$ can only reduce the number of mistakes, so the same lower bound still follows. Exact probabilistic dimension uses a full-support marginal; zero loss means that every point is correct. These observations make the conclusions insensitive to either fixed tie rule, without weakening the universal quantifiers over D and h .

B A finite-bit convex program for the cap implementation

This appendix proves the weakly polynomial implementation needed in Theorem 16. It does not require a claim that an exact radial-isotropic matrix has rational entries. Approximate Forster computation is established prior work; see Diakonikolas et al. [12] for stronger algorithms. We verify the particular general-position convex program and the bit bounds needed here.

Let $k > d$ and let $v_1, \dots, v_k \in \mathbb{Q}^d$ be in linear general position. Assume the numerator and denominator of every coordinate have absolute value at most 2^b , with nonzero denominators. Let Δ_I be the determinant of the rows indexed by a d -subset I . Every Δ_I is nonzero. Cauchy–Binet gives the convex function

$$f(t) = \log \det \left(\sum_i e^{t_i} v_i v_i^\top \right) - \frac{d}{k} \sum_i t_i = \log \sum_{|I|=d} \Delta_I^2 e^{\sum_{i \in I} t_i} - \frac{d}{k} \sum_i t_i. \quad (21)$$

The identity follows by applying the determinant expansion to the product of the $d \times k$ column matrix with entries $e^{t_i/2} v_i$ and its transpose; only distinct d -subsets remain. Differentiating the log-sum-exp expression shows that its Hessian is the covariance matrix of the incidence vector $\mathbf{1}_I$ under the displayed positive weights. It is positive semidefinite and has operator norm at most $\mathbb{E} \|\mathbf{1}_I\|^2 = d$. Also $\|\nabla f\| \leq 2\sqrt{d}$. Thus f is convex, d -smooth, and invariant under adding a constant to all coordinates.

We give a polynomial radius bound on the zero-sum gauge. Write $a_- = \min_I \log \Delta_I^2$ and $a_+ = \max_I \log \Delta_I^2$. For $t_1 \geq \dots \geq t_k$ with $\sum t_i = 0$, put $s = t_1 + \dots + t_d$. The average of the largest $d - 1$ of $t_2, \dots, t_k$ is at least their overall average; hence

$$s \geq t_1 \frac{k - d}{k - 1} \geq \frac{t_1}{k - 1}.$$

Likewise the largest d of $t_1, \dots, t_{k-1}$ have average at least $-t_k/(k - 1)$, giving $s \geq -dt_k/(k - 1)$. Therefore $s \geq \|t\|_\infty/(k - 1)$, and

$$f(t) \geq a_- + \frac{\|t\|_\infty}{k - 1}, \quad f(0) \leq a_+ + \log \binom{k}{d}.$$

Sublevel sets on the gauge are compact. A minimizer exists and has infinity norm at most $(k - 1)(a_+ - a_- + \log \binom{k}{d})$.

The determinant bounds are explicit. A common denominator for a $d \times d$ determinant is the product of its d^2 coordinate denominators, so

$$2^{-bd^2} \leq |\Delta_I| \leq d! 2^{bd}.$$

Consequently the integer

$$R = 1 + (k - 1)(2bd^2 + 2bd + 2d\lceil \log_2(d + 1) \rceil + d\lceil \log_2(k + 1) \rceil)$$

exceeds the minimizer radius. Work on the rational polytope $Q = \{t : \sum t_i = 0, |t_i| \leq 2R\}$.

For completeness, a deliberately conservative projected-subgradient implementation suffices. Set

$$\delta = \frac{d}{512k^2}, \quad a = \frac{\delta}{36d}, \quad \xi = \frac{\delta}{32Rk}, \quad T = \left\lceil \frac{4R^2k}{a\delta} \right\rceil.$$

At each step compute a rational approximation $\tilde{g}$ of ∇f with Euclidean error at most ξ , and project $t - a\tilde{g}$ onto Q . The projection is rational and polynomial-time: it is $\text{clip}(z_i - \lambda, -2R, 2R)$ in each coordinate, with λ chosen to make the sum zero. The sum is a continuous piecewise-linear decreasing function; sorting its $2k$ breakpoints and solving on the containing interval finds λ . The projection inequality follows directly from its normal-cone signs: for every $u \in Q$, $\langle z - \Pi_Q z, u - \Pi_Q z \rangle \leq 0$, so $\|\Pi_Q z - u\| \leq \|z - u\|$.

Let t^* be a minimizer with $\|t^*\| \leq R\sqrt{k}$, and start at zero. The standard distance expansion, supplied here to account for error, gives

$$\frac{1}{T} \sum_{s < T} (f(t_s) - f(t^*)) \leq \frac{R^2k}{2aT} + \frac{a}{2}(2\sqrt{d} + \xi)^2 + 4R\sqrt{k}\xi.$$

Indeed convexity bounds $f(t_s) - f(t^*)$ by $\langle \nabla f(t_s), t_s - t^* \rangle$, the gradient error contributes at most $4R\sqrt{k}\xi$, and squared distances telescope. Since $\xi \leq \sqrt{d}$, each of the three terms is at most $\delta/8$. Convexity gives $f(\bar{t}) - \min f < \delta$ for the rational average $\bar{t}$.

Smoothness on the full space implies

$$\|\nabla f(\bar{t})\|^2 \leq 2d(f(\bar{t}) - \min f) < \frac{d^2}{256k^2} :$$

apply the quadratic upper bound to $\bar{t} - \nabla f(\bar{t})/d$ and compare with the global minimum. Put $S = \sum_i e^{\bar{t}_i} v_i v_i^\top$. Differentiating the first expression in (21) yields

$$\frac{15d}{16k} < \ell_i := e^{\bar{t}_i} v_i^\top S^{-1} v_i < \frac{17d}{16k}.$$

For any factor T_0 with $T_0 S T_0^\top = I$, the normalized row covariance satisfies

$$\frac{16}{17d} I \preceq \frac{1}{k} \sum_i \frac{T_0 v_i v_i^\top T_0^\top}{\|T_0 v_i\|^2} \preceq \frac{16}{15d} I.$$

To see this, write each summand as $e^{\bar{t}_i} T_0 v_i v_i^\top T_0^\top / \ell_i$ and use the preceding bounds together with their unnormalized sum I .

All calculations can be performed to the required accuracy with polynomially many bits, rather than an unspecified exact-real oracle. For fixed d , the number of terms $\binom{k}{d}$ in (21) is polynomial. The gradient is a ratio of sums of the nonnegative numbers $\Delta_I^2 \exp(\sum_{i \in I} t_i)$. Their exponent arguments have absolute value at most $2dR$, and their rational coefficients have polynomial bit length by the determinant bounds. Exponentials on this bounded range admit rational interval approximations with bit complexity polynomial in $R, b, k, \log(1/\xi)$: for nonnegative arguments use the Taylor series and its positive tail bound, and for negative arguments take reciprocals. The elementary inequality $j! \geq (j/e)^j$, obtained by integrating $\log x$, bounds the number of Taylor terms by a polynomial in the argument bound and requested precision. Positive sums have no cancellation; the denominator has the explicit lower bound $2^{-2bd^2} e^{-2dR}$. Computing each gradient coordinate to error at most ξ/k gives the required Euclidean error. Exact rational projection and averaging increase bit lengths by

only a polynomial over the polynomial number of iterations. Thus the preceding inexact-gradient implementation has a polynomial bit-operation bound for fixed d .

Finally, approximate T_0 by an invertible rational matrix T . This also requires only polynomial precision. The bounds

$$\lambda_{\max}(V^\top V) \leq kd2^{2b}, \quad \lambda_{\min}(V^\top V) \geq \frac{2^{-2bd^2}}{(kd2^{2b})^{d-1}}$$

follow from trace, determinant, and Cauchy–Binet. Since $|\bar{t}_i| \leq 2R$, multiplying these bounds by $e^{\pm 2R}$ bounds the eigenvalues of S . Their logarithms have polynomial size. Interval Cholesky factorization followed by inversion, approximating each square root by rational bisection, consequently computes T_0 to $2^{-q(k,b,R)}$ operator error in polynomial time for a sufficiently large fixed polynomial q . This last assertion can also be checked directly: Cholesky pivots are bounded below by the smallest eigenvalue, all divisions have polynomial logarithmic conditioning, and there are only $O(d^3)$ operations. If $\|a - b\| \leq \|a\|/2$, then

$$\|a/\|a\| - b/\|b\|\| \leq 2\|a - b\|/\|a\|.$$

The displayed eigenvalue bounds and the nonzero rational input coordinates therefore give an explicit polynomial precision at which the normalized covariance changes by less than $1/(8d)$. Its lower eigenvalue remains above $3/(4d)$. For rational T , this final statement is independently decidable by rational arithmetic: the covariance is the sum of $Tv_i v_i^\top T^\top / (v_i^\top T^\top T v_i)$, and positivity above $3I/(4d)$ is checked using leading principal minors. This completes the finite-bit construction.

SQ replies themselves need no unlimited precision in the computational corollary. Request half the advertised tolerance and round the reply to a rational grid of spacing at most that tolerance. The effective error remains within the advertised tolerance. This changes only an absolute constant and uses $O(\log(1/\tau))$ reply bits. Each cap test is a fixed-degree rational inequality; the unpeeled set is represented by the negations of earlier peeling tests. No table-dependent advice beyond the stated input is hidden in this computation.

C The cube concentration calculation used for non-necessity

Let f on the uniform n -cube change by at most one when one coordinate is flipped. Reveal the coordinates successively and write the Doob differences as D_i . Conditional on the first $i - 1$ coordinates, the two conditional expectations after revealing coordinate i differ by at most one: couple the remaining coordinates identically and use the bounded-difference hypothesis. Thus D_i is conditionally centered and lies in an interval of length at most one.

For any centered random variable Z in an interval of length one, let $F(\lambda) = \log \mathbb{E}e^{\lambda Z}$. Differentiation of the finite sum gives $F''(\lambda)$ equal to the variance under the exponentially tilted law. A variable $Y \in [a, b]$ has variance at most $(b - a)^2/4$: from $(Y - a)(b - Y) \geq 0$ obtain $\text{Var}(Y) \leq (\mathbb{E}Y - a)(b - \mathbb{E}Y) \leq (b - a)^2/4$. Hence $F'' \leq 1/4$, and integrating twice using $F(0) = F'(0) = 0$ yields $F(\lambda) \leq \lambda^2/8$ for either sign of λ .

Iterating conditional expectation gives

$$\mathbb{E} \exp\{\lambda(f - \mathbb{E}f)\} \leq \exp(n\lambda^2/8).$$

Markov's inequality and $\lambda = 4a/n$ imply $\Pr[f - \mathbb{E}f \geq a] \leq e^{-2a^2/n}$. Apply the same calculation to $-f$ for the lower tail. All distributions here are finite, so differentiation and conditional expectations require no limiting justification.

D CPU verification and adversarial oracle experiments

All new experiments are CPU-only. They test the cap-based learner and its proper completion, not SGD. The final reporting seed is 2026091103; the separate algebraic-check seed is 2026091107. The scripts, input families, all rational distributions, every query and answer, numerical diagnostics, exact finite-coverage certificates, and independent audit outputs are included in the accompanying archive. Execution used Python 3.13.5, NumPy 2.3.5, SciPy, mpmath, and one numerical thread. The final learner experiment took 300.18 seconds in this environment; this observed runtime is not a complexity theorem.

Executed cap law versus the ideal sampler. The practical implementation perturbs the template once, computes numerical radial-isotropic transforms with the convex log-determinant objective, and constructs a finite pool from point-centered directions and 32 seeded Gaussian directions. This pool is not the ideal continuous uniform-sphere law and is not the enormous deterministic net used in the bit-complexity proof. The numerical solver uses L-BFGS, with a 120-decimal Newton fallback and a separate small-survivor formula. Early floating-point conditioning failures are retained as diagnostic logs. No failed numerical transform was silently counted as a verified transform.

The guarantee used for each reported state is instead *the exact finite coverage of the actual pool*. The script checks each resulting rectangle against the integer table and verifies the pointwise coverage inequality by integer counting. Conditioning the pool to remove known-empty point sets can only increase coverage. The catalog depends only on the survivor set and a fixed seed, not on the unknown marginal or on previous instances. Across 1,735 catalogs, all 1,106,791 rectangles were monochromatic, and all 1,096,914 pointwise constraints passed. The minimum certified coverage was 0.00013950892857142856, above the common required bound 2^{-20} . The largest numerical covariance residual was $2.003 \cdot 10^{-5}$. Exact coverage, not this floating-point residual, certifies application of the learner’s argument to each recorded catalog.

This is a certificate-checked implementation on the reported finite states, not a verified polynomial-time numerical solver on arbitrary input. Appendix B proves a separate conservative bit-polynomial algorithm. We do not describe an approximate matrix as an exact irrational optimizer.

Oracle policy. The exact oracle computes expectations using rational arithmetic. The adversarial policy always chooses a tolerance endpoint and uses the complete *public* query-answer history. On a proposed rectangle’s marginal-mass query it returns the lower endpoint to obstruct acceptance; it does not inspect the private row side or the as-yet-unrevealed color. On a residual-mass query it prefers an available premature stop with larger current error, and otherwise delays stopping. A mismatch query itself reveals its color; the policy chooses an endpoint that induces a wrong peel when possible, otherwise preferring elimination. On candidate verification it favors a high-error acceptance when available. In the proper completion it favors the small-tail branch when the endpoint permits it. The exact tie rules and decision functions are in the script and are replayed by the independent audit.

This policy is greedy with respect to immediate error and continuation. It is explicitly adversarial, but it is not proved to maximize final error or runtime over all possible oracle policies and future learner coins. In particular, the marginal answer does not read private T or c information that the query has not disclosed. An earlier diagnostic version did so; its outputs were discarded and the final experiments were rerun with the public-transcript policy.

Distribution search. For every $N = 2, \dots, 8$, the script generates a fixed random flip matrix and uses 16 reporting distributions, each supported on at most 20 points. Structured cases include mass on a target’s template incidences, on its actual negative points, on three points, and on a small tail just above the stopping threshold. Four random restarts with three coordinate mutations each search for larger error and round count, with proper and improper objectives included. Learner reporting seeds are separate from the pilot seeds. Every case is run with both learners and both oracles, giving $7 \cdot 16 \cdot 2 \cdot 2 = 448$ runs.

The outer accuracy is $\epsilon = 1/10$. To compare the variants on the same tolerance, both run the rectangle phase at $\theta = 1/40$, with

$$\rho = 2^{20}, \quad P_\theta = 24194093, \quad \tau = \frac{1}{23226329280}, \quad R = 151681296.$$

These conservative universal budgets greatly exceed observed use. The table reports maxima over both exact and adversarial-oracle runs, not certified worst cases over all distributions or policies.

N	Improper cap learner			Proper cap learner		
	Error	Rounds	Queries	Error	Rounds	Queries
2	0.0062500001	78	163	0.0062500001	73	152
3	0.0062500001	268	543	0.0099348873	268	544
4	0.0062500001	667	1351	0.0062500001	565	1139
5	0.0062500001	3845	7701	0.0062500001	3845	7702
6	0.0062500001	2038	4086	0.0062500001	2038	4087
7	0.0062500001	7227	14464	0.0062500001	7227	14465
8	0.0062500001	4190	8392	0.0062500001	4190	8393

Error entries are rounded upward; raw rational errors are retained. The largest actual improper error was 0.006250000021527293 and the largest proper error was 0.009934887239. Exact-oracle trajectories sometimes take more rounds than the greedy endpoint policy: an adversarial premature stop raises error but can shorten the run. This is not a contradiction and is why runtime and error are reported separately.

The audit checked 339,922 query-answer pairs, including 103,989 endpoint answers, and all 224 proper outputs belonged to the supplied class. It reconstructed the residual set, candidate set, partial predictor, target-preservation invariant, and final error from each transcript. Proposal outcomes totaled 1,432 negative-or-positive peel decisions, 2,732 eliminations, and 163,462 rejections. A negative peel decision in the proper learner interrupts peeling and starts proper completion; it is not necessarily committed to the partial predictor. Termination counts were 279 residual stops, 105 rectangle-candidate acceptances, 28 identified rows, 28 small-tail candidate acceptances, and eight empty residual sets. All invariant and oracle checks passed.

Product-distribution rectangle witnesses. To inspect the mechanism rather than assume uniform product weights, a separate search uses ten restarts, four support-coordinate changes, and five alternating linear-program cycles on at most ten rows and ten columns. The resulting weights are rounded to rational masses with denominator 10^8 . The largest monochromatic rectangle under each resulting product measure is then computed exactly by exhausting all row subsets and both colors. This gives a rigorous *upper bound* on the infimum defining $\text{rect}(A)$, not an exact computation of that infimum. Together with Theorem 11, the certified intervals are

$$2^{-15} \leq \text{rect}(A) \leq u_N,$$

where the following decimal upper bounds are rounded upward.

N	2	3	4	5	6	7	8
u_N	.242215	.222223	.243056	.244898	.244898	.244083	.244898

Zero-weight selected rows and columns are allowed. The enumeration still certifies the complete product measure because all unselected entries have zero mass. All seven witnesses passed the independent exact audit. The intervals are wide; the experiment does not claim a numerical characterization of rect or asymptotic behavior from $N \leq 8$.

Finite inequality checks. The auxiliary script checked 80,000 combinations of extreme valid answers for the acceptance inequalities, 2,000 elimination sequences, 500 exact sign-strictification examples, 825 exact star-union identification identities, exhaustive uniform-cube rectangles for $n = 1, 3$, and 400 arbitrary-product projective-plane bounds. It verified the incidence and empty-row counts for all seven grid sizes. The empty-row counts were 3, 6, 10, 15, 21, 28, 36; the incidence counts were 23, 126, 412, 1025, 2151, 4018, 6896. Every assertion passed. These are finite checks of the formulas, not substitutes for their universal proofs.

Scope of verification. The paper supplies human-readable proofs of its new universal statements. The numerical searches do not certify global adversaries or optimizers. The practical solver’s exact catalog checks certify its recorded states, not all possible states on every possible input. Warren’s sign-pattern theorem, the sharper homogeneous-rectangle theorem, and real-closed-field decidability are imported classical results with explicit citations rather than new complete formalizations. The pinned Lean sources are uncompiled. Open questions are labeled as such, and no empirical observation is promoted to an unrestricted theorem.

E Finite Lean targets and verification status

The definitions of finite distributions, strict dimension, SQ queries, adaptive real-response decision trees, oracle validity, and distribution-independent randomized learning remain in `SQDC.lean`. Four additional finite statements are in `Turn2.lean`. The Lean sources encode labels as `Bool`, using `true` for $+1$ and `false` for -1 . They are stated here to make the source-to-paper correspondence exact.

Lemma L1 (Peeling error). *Let $o : X \rightarrow \{\perp\} \cup [t]$ assign every point to at most one peel. Let $c_i \in \{-1, +1\}$ and let g_0 be any fallback. Define $g(x) = c_i$ if $o(x) = i$, and $g(x) = g_0(x)$ otherwise. Put $w_i = D\{x : o(x) = i, c_i \neq h(x)\}$ and $u = D\{x : o(x) = \perp\}$. Then $L_{D,h}(g) \leq \sum_i w_i + u$. If $w_i \leq 3\tau$ and $u \leq u_0$, then $L_{D,h}(g) \leq 3t\tau + u_0$.*

Proof. Partition the finite sum defining loss by $o(x)$. On a peeled set its contribution is exactly w_i ; on the residual set each error indicator is at most one. Sum the pointwise inequalities and then the upper bounds on w_i . $\square$

Lemma L2 (Elimination potential). *Let $p_0, \dots, p_R > 0$, $p_R \leq 1$, and $0 \leq b_t < 1$ with $p_{t+1} = p_t/(1 - b_t)$ for $0 \leq t < R$. Then $\sum_{t < R} b_t \leq -\ln p_0$.*

Proof. The inequality $\ln(1 - b_t) \leq -b_t$ gives $b_t \leq \ln p_{t+1} - \ln p_t$. Sum and use $\ln p_R \leq 0$. $\square$

Lemma L3 (Positive rectangles survive). *If $F_{ij} = +1$ implies $A_{ij} = +1$ for every entry, every positive rectangle of F is a positive rectangle of A .*

Proof. Apply the entrywise implication at every pair in the rectangle. $\square$

Lemma L4 (Two integer grid lines). *Let $a, b, c, d, x, y, x', y' \in \mathbb{Z}$ with $(a, b) \neq (c, d)$, $y = ax + b = cx + d$, and $y' = ax' + b = cx' + d$. Then $x = x'$ and $y = y'$.*

Proof. If $a = c$, the first equality gives $b = d$, a contradiction. Subtracting the two shared-point equalities gives $(a - c)(x - x') = 0$. The integers have no zero divisors, so $x = x'$, and substitution gives $y = y'$. $\square$

The Lean scripts are unverified. In this turn, `lake -version` returned `command-not-found` and exit 127. No further compiler or installation effort was attempted. The previous turn also recorded unavailable build commands. The source is pinned to Mathlib commit

2631d1cc8c2ace6c6a900425d6e5d2b5963966e9

, whose toolchain file specifies `leanprover/lean4:v4.34.0-rc2`. The repository commit and toolchain were read through the GitHub connector. The scripts contain no deliberately inserted `sorry` or custom axioms, but no successful compilation or axiom audit is claimed. The main randomized learner and the real-algebraic counting theorem are not yet formalized. Build instructions and actual failure logs are included.

F Retained independent geometric bounds

Proposition 14 (Independent constant rectangle bound). *Let c_d be the normalized surface measure of the preceding cap on S^{d-1} , $d \geq 2$. Every matrix F of sign-rank at most d satisfies $\text{rect}(F) \geq c_d/(4d)$. If F has no negative 2×2 , every monotone flip A satisfies $\text{rect}(A) \geq c_d/(8d)$. In particular, for $d = 6$ this is greater than 2^{-18} .*

Proof. First take a uniform prior on $k > d$ row copies. Perturb their representing vectors, if necessary, so every d are independent while preserving the finitely many strict signs. The full minimization argument in retained Theorem 3 produces an invertible transform whose normalized row vectors are isotropic. Apply the inverse transpose to the point vectors and normalize; all signs persist.

For each uniform random direction u , the two sets $T_+(u) = \{v : \langle u, v \rangle > \alpha\}$ and $T_-(u) = \{v : \langle u, v \rangle < -\alpha\}$ have combined prior mass at least $1/(2d)$. The rectangles $C_u \times T_+(u)$ and $C_u \times T_-(u)$ have constant signs. For any fixed point marginal μ , rotational invariance gives $\mathbb{E}_u \mu(C_u) = c_d$, because each unit point lies in a random cap with probability c_d . Therefore the expected sum of their product masses is at least $c_d/(2d)$, and some single rectangle has mass at least $c_d/(4d)$.

For a rational row prior, duplicate rows with common-denominator multiplicities, repeating all copies further to ensure $k > d$. The transformed representation may differ across identical copies, but every pairwise sign still equals the original matrix entry. Projecting a rectangle to original rows only increases its prior mass. Approximation by rational priors and continuity of the finite maximum prove the assertion for all priors. A negative rectangle has a singleton side under the additional assumption on F , so the same half-mass argument as in (12) proves the second assertion.

For the numerical estimate at $d = 6$, project the cap onto its last five coordinates. Its image is the five-dimensional ball of radius $\alpha = 1/\sqrt{12}$, and the surface-area Jacobian is at least one. The volume of the unit five-ball is $8\pi^2/15$ and the area of S^5 is π^3 , obtained by polar integration of the Gaussian integral. Thus

$$c_6 \geq \frac{8}{15\pi^{12^{5/2}}} > \frac{8}{15 \cdot 4 \cdot 576} = \frac{1}{4320}, \quad \frac{c_6}{48} > \frac{1}{207360} > 2^{-18}.$$

This independent constant, rather than the sharper imported one, is already sufficient for Theorems 9–12 to refute every polynomial dimension implication. $\square$

Proposition 15 (Uniform-product rectangles in a projective plane). *For the point–line incidence sign matrix of $\text{PG}(2, q)$, let $K = q^2 + q + 1$ and $k = q + 1$. Under uniform distributions on both sides, the largest monochromatic-rectangle mass lies between*

$$\frac{1}{8(q+1)} \quad \text{and} \quad \frac{q}{(q+1)^2}.$$

It is therefore $\Theta(1/q)$.

Proof. Let P be the incidence matrix. As proved in retained Theorem 4, $P\mathbf{1} = k\mathbf{1}$ and $P^T P = qI + \mathbf{1}\mathbf{1}^T$. Thus $C = P - (k/K)\mathbf{1}\mathbf{1}^T$ has operator norm $\sqrt{q}$. A nonincident rectangle with side sizes a, b satisfies

$$\frac{k}{K}ab = |\mathbf{1}_S^T C \mathbf{1}_T| \leq \sqrt{q}\sqrt{ab},$$

so its mass is at most q/k^2 . An all-incident rectangle has a singleton side and size at most k . Its mass k/K^2 is also at most q/k^2 : use $K \geq qk$ and $q^3 \geq k$ for $q \geq 2$ to obtain $qK^2 \geq q^3k^2 \geq k^3$.

For the lower bound, choose $s = \lfloor K/(2k) \rfloor$ lines. Their union has at most $sk \leq K/2$ points, leaving at least $K/2$ points incident with none of them. The resulting rectangle has mass at least $s/(2K)$. Since $K/(2k) \geq 1$, $s \geq K/(4k)$, proving the lower bound. $\square$

G Retained barrier, plane, and transcript results

Lemma 1 (Finite energy and threshold counting). *Let P_0 be a probability distribution on a finite set Z , and let $r_1, \dots, r_K \in L^2(P_0)$. Set*

$$\Gamma_{ij} = \mathbb{E}_{P_0} r_i r_j, \quad \Lambda = \|\Gamma\|_{\text{op}}.$$

For every $a : Z \rightarrow \mathbb{R}$,

$$\sum_{j=1}^K (\mathbb{E}_{P_0} r_j a)^2 \leq \Lambda \mathbb{E}_{P_0} a^2. \quad (22)$$

In particular, if $|a| \leq 1$ and $\tau > 0$, then

$$|\{j : |\mathbb{E}_{P_0} r_j a| > \tau\}| \tau^2 \leq \Lambda. \quad (23)$$

More generally, for any finite list of real numbers c_i , $\tau \geq 0$, and $\sum_i c_i^2 \leq \Lambda$, one has $|\{i : \tau < |c_i|\}| \tau^2 \leq \Lambda$.

Proof. Let $T : \mathbb{R}^K \rightarrow L^2(P_0)$ send b to $\sum_j b_j r_j$. Then $T^* T = \Gamma$. For every unit vector b ,

$$|\langle b, T^* a \rangle| = |\langle Tb, a \rangle| \leq \|Tb\|_2 \|a\|_2 \leq \sqrt{\Lambda} \|a\|_2.$$

Taking the supremum over b and squaring proves (22). Each selected summand in (23) is greater than τ^2 ; discarding nonnegative unselected summands proves the result. The same argument, with weak inequalities, proves the final statement even when $\tau = 0$. $\square$

Theorem 2 (Joint-distribution SQ barrier). *Let μ be a full-support distribution on X and let $P_0 = \mu \otimes \text{Unif}\{-1, +1\}$. For $j \in [K]$, let P_j be the labeled distribution arising from some D_j and $h_j \in \mathcal{H}$. Put*

$$r_j = \frac{P_j}{P_0} - 1, \quad \Gamma_{ij} = \mathbb{E}_{P_0} r_i r_j, \quad \Lambda = \|\Gamma\|_{\text{op}}, \quad R = K/\Lambda.$$

Any randomized adaptive (m, τ) -SQ learner satisfying (1) obeys

$$1 - 2\epsilon \leq \sqrt{\frac{\Lambda}{K}} + \frac{m\Lambda}{K\tau^2}, \quad \text{equivalently} \quad \frac{m}{\tau^2} \geq (1 - 2\epsilon)R - \sqrt{R}. \quad (24)$$

Consequently, for $\epsilon < 1/4$,

$$R \leq 4 + 4m/\tau^2. \quad (25)$$

Proof. First $\Lambda > 0$. Indeed, with $a_j(x, y) = yh_j(x)$ we have $\|a_j\|_{L^2(P_0)} = 1$ and $\langle r_j, a_j \rangle = \mathbb{E}_{P_j} a_j - \mathbb{E}_{P_0} a_j = 1$. Thus $\|r_j\|_2 \geq 1$ and $\Lambda \geq \Gamma_{jj} \geq 1$.

For a fixed private seed, run the learner on the reference responses $\mathbb{E}_{P_0} q$. Write the resulting queries as $q_1, \dots, q_m$ and the leaf output as g_0 . This is a counterfactual tree path; the reference need not be realizable. For target j , choose the following memoryless oracle, which does not know the private seed:

$$O_j(q) = \begin{cases} \mathbb{E}_{P_0} q, & |\mathbb{E}_{P_j} q - \mathbb{E}_{P_0} q| \leq \tau, \\ \mathbb{E}_{P_j} q, & \text{otherwise.} \end{cases}$$

It is valid for P_j on every query. If its run first departs from the reference transcript, that departure occurs at one of the reference queries with $|\langle r_j, q_t \rangle| > \tau$. Lemma 1 and a union bound give, for this fixed seed,

$$\frac{1}{K} |\{j : \text{the transcript departs from the reference}\}| \leq \frac{m\Lambda}{K\tau^2}.$$

For the reference leaf, define $c_j = \mathbb{E}_{P_j}[yg_0(x)]$. Since $\mathbb{E}_{P_0}[yg_0(x)] = 0$, Lemma 1 gives

$$\frac{1}{K} \sum_j |c_j| \leq \sqrt{\frac{1}{K} \sum_j c_j^2} \leq \sqrt{\Lambda/K}.$$

Let s_j be the correlation of the actual output under O_j . On a nondeparting transcript it equals c_j ; on a departing transcript it is at most 1. Therefore $s_j \leq |c_j| + \mathbf{1}_{\text{departure for } j}$. Average this inequality over j and then over the private seed. The learner's guarantee gives an average correlation at least $1 - 2\epsilon$. This proves (24). Notice that we did not choose a different oracle after seeing the seed.

Finally write $M = m/\tau^2$. From (24), $R/2 \leq \sqrt{R} + M$. The square $(\sqrt{R} - 2)^2 \geq 0$ gives $\sqrt{R} \leq R/4 + 1$. Combining yields (25). $\square$

The requested fixed-marginal barrier. Set $D_j = \mu$ for every j . Then $r_j(x, y) = yh_j(x)$ and

$$\Gamma_{ij} = G_{ij} = \mathbb{E}_\mu[h_i h_j].$$

For $R = K/\|G\|_{\text{op}} \geq 16$, (24) implies

$$m \geq \frac{1}{4} R \tau^2. \quad (26)$$

Indeed, $1 - 2\epsilon > 1/2$ and $\sqrt{R} \leq R/4$. The cutoff matters: a uniformly positive lower bound proportional to $R\tau^2$ cannot hold for all tiny families, since a singleton needs no queries. The next theorem supplies the accompanying dimension lower bound.

G.1 The retained sign-rank lower bound

We use the spectral method associated with Forster [7], but include its geometric step rather than treating the published theorem as a proof.

Theorem 3 (Spectral sign-rank bounds). *If $A \in \mathbb{R}^{N \times K}$ satisfies $|A_{xj}| \geq 1$ for all entries, then*

$$\text{signrank}(\text{sign } A) \geq \frac{\sqrt{NK}}{\|A\|_{\text{op}}}. \quad (27)$$

If $M \in \{-1, +1\}^{N \times K}$ and $G = M^\top \text{diag}(\mu)M$ for any distribution μ on its rows, then

$$\text{signrank}(M) \geq \sqrt{K/\|G\|_{\text{op}}}. \quad (28)$$

Proof. Let $d = \text{signrank}(M)$, with $M = \text{sign } A$ in the first assertion. Choose a strict representation $M_{xj} = \text{sign}\langle z_x, v_j \rangle$ in $\mathbb{R}^d$. If $d \geq K$, both conclusions are immediate: $\|A\|_{\text{op}} \geq \sqrt{N}$ from any column, while $G_{jj} = 1$ implies $\|G\|_{\text{op}} \geq 1$. We henceforth have $1 \leq d < K$.

Perturb the finitely many v_j within sufficiently small open neighborhoods so that signs are unchanged and every d of them are independent. Such a perturbation exists: for each d -subset the determinant is a nonzero polynomial in the vector coordinates, and a finite union of its zero sets has empty interior. To justify this elementary fact, a nonzero univariate polynomial has finitely many roots; induction on the number of variables, by viewing the polynomial as a polynomial in the last variable, shows that a multivariate polynomial cannot vanish on a nonempty open box. Apply this to the product of the finitely many determinant polynomials.

We prove that there is an invertible T such that the normalized vectors $u_j = Tv_j/\|Tv_j\|$ satisfy

$$\sum_{j=1}^K u_j u_j^\top = (K/d)I_d. \quad (29)$$

For every $1 \leq r \leq d$, there is a common $\delta > 0$ such that at most $d - r$ of the v_j have projection norm less than δ onto any r -dimensional subspace. In fact, any $d - r + 1$ of the vectors are independent, so cannot all belong to the orthogonal complement of an r -dimensional subspace. Their maximum projection norm is therefore positive for every such subspace. Its minimum is positive by compactness of the space of orthogonal projection matrices of rank r ; this space is closed and bounded. Take the minimum over finitely many subsets and r .

For positive definite Q with $\det Q = 1$, minimize

$$F(Q) = \sum_{j=1}^K \log(v_j^\top Q v_j).$$

Write the decreasing eigenvalues of Q as $e^{s_1} \geq \dots \geq e^{s_d}$, where $\sum_i s_i = 0$. For each r , at least $K - d + r$ of the quadratic forms are at least $\delta^2 e^{s_r}$. Order the $K \log$ quadratic forms increasingly. For the first d entries use respectively the bounds $\log \delta^2 + s_d, \dots, \log \delta^2 + s_1$; each of the remaining $K - d$ entries is at least $\log \delta^2 + s_1$. Thus

$$F(Q) \geq K \log \delta^2 + (K - d)s_1.$$

Since $K > d$, every sublevel set has bounded largest eigenvalue. The determinant condition then bounds the smallest eigenvalue away from zero. Sublevel sets are compact, so a minimizer Q exists.

For a symmetric traceless matrix B , the path $Q(t) = Q^{1/2} \exp(tB) Q^{1/2}$ retains determinant one. Its derivative at zero gives

$$0 = \sum_j \frac{v_j^\top Q^{1/2} B Q^{1/2} v_j}{v_j^\top Q v_j} = \text{tr} \left(B \sum_j u_j u_j^\top \right), \quad u_j = \frac{Q^{1/2} v_j}{\|Q^{1/2} v_j\|}.$$

A symmetric matrix orthogonal to every traceless symmetric matrix is scalar: subtract its scalar trace part and take that difference as B . Taking traces gives (29), with $T = Q^{1/2}$.

Replace each row vector by the normalization of $T^{-\top} z_x$, denoted e_x , so $\|e_x\| = 1$ and all signs are preserved. Let U be the matrix with columns u_j . For every row of A ,

$$\langle e_x, U A_{x,\cdot}^\top \rangle = \sum_j |A_{xj}| |\langle e_x, u_j \rangle| \geq \sum_j \langle e_x, u_j \rangle^2 = K/d.$$

Here $|\langle e_x, u_j \rangle| \leq 1$ was used. Consequently,

$$NK^2/d^2 \leq \|U A^\top\|_F^2 \leq \|A\|_{\text{op}}^2 \|U\|_F^2 = K \|A\|_{\text{op}}^2.$$

This proves (27). For M , use the same row inequality, square, and average under μ :

$$K^2/d^2 \leq \sum_x \mu(x) \|U M_{x,\cdot}^\top\|^2 = \text{tr}(U G U^\top) \leq \|G\|_{\text{op}} \text{tr}(U U^\top) = K \|G\|_{\text{op}}.$$

The last inequality follows by diagonalizing the positive semidefinite G , whose eigenvalues lie between zero and its operator norm. This proves (28). $\square$

Theorems 2 and 3 prove the valid conditional assertion in the proposed spectral barrier: a certificate with ratio R yields dimension lower bound $\sqrt{R}$ and query lower bound $\Omega(R\tau^2)$ when R is large. They do not prove that all explicit sign-rank lower bounds admit such an ordinary label-Gram certificate. The following family disproves that extra assertion in precisely this form.

G.2 Projective planes

Theorem 4 (An escaping plain-Gram family, but not an SQ separation). *For each prime $q \geq 2$, let X_q be the points of the projective plane over $\mathbb{F}_q$, and let $\mathcal{H}_q$ have one hypothesis per line, positive on that line and negative elsewhere. Set $N = q^2 + q + 1$ and $k = q + 1$. Then:*

$$\frac{q+1}{\sqrt{q}} \leq \text{dc}(\mathcal{H}_q) \leq 2q+3, \quad (30)$$

$$\frac{|J|}{\|[\mathbb{E}_\mu h_i h_j]_{i,j \in J}\|_{\text{op}}} \leq 16 \quad \text{for every distribution } \mu \text{ and nonempty } J \subseteq \mathcal{H}_q. \quad (31)$$

There is a distribution-independent N -query tolerance- τ learner of loss at most τ . For each fixed D and $0 < \epsilon < 1/4$, there is a learner allowed to depend on D , with tolerance $\epsilon/2$ and

$$m_D \leq 1 + s + \binom{s}{2}, \quad s = \left\lceil \frac{4 \log N + 2}{\epsilon} \right\rceil,$$

that has loss at most ϵ for every target. Here $\log$ is natural. Nevertheless every single learner satisfying (1), with $m \geq 1$, $0 < \tau \leq 1$, and $\epsilon < 1/4$, satisfies

$$\text{dc}(\mathcal{H}_q) \leq 19m/\tau^2. \quad (32)$$

Proof. Points are one-dimensional subspaces of $\mathbb{F}_q^3$; lines are kernels of nonzero linear functionals, modulo nonzero scalar multiples. There are $(q^3 - 1)/(q - 1) = N$ of each. A line contains $(q^2 - 1)/(q - 1) = k$ points. Two distinct points span a unique two-dimensional subspace, hence lie on one common line. Two distinct kernels intersect in one one-dimensional subspace. These facts also show each point lies on k lines.

Let P be the $N \times N$ incidence matrix. Then

$$P\mathbf{1} = k\mathbf{1}, \quad P^\top P = qI + \mathbf{1}\mathbf{1}^\top.$$

Define $A = (N/k)P - \mathbf{1}\mathbf{1}^\top$. Its positive entries equal $N/k - 1 = q^2/(q + 1) \geq 1$, and its negative entries equal -1 . Direct multiplication gives

$$A\mathbf{1} = 0, \quad A^\top A = (N/k)^2 q \left(I - \frac{\mathbf{1}\mathbf{1}^\top}{N} \right).$$

Indeed, the coefficient of $\mathbf{1}\mathbf{1}^\top$ before simplification is $N^2/k^2 - N = -Nq/k^2$, since $k^2 = N + q$. Thus $\|A\|_{\text{op}} = (N/k)\sqrt{q}$, and Theorem 3 gives the lower bound in (30).

For the upper bound, enumerate the points by distinct integers $1, \dots, N$ and use the common embedding $\phi(t) = (1, t, \dots, t^{2k})$. For a positive set S of size k , the polynomial

$$p_S(t) = \frac{1}{2} - \prod_{a \in S} (t - a)^2$$

is positive on S and negative on all other domain points: away from S the product is a positive integer. Its coefficient vector is the required weight vector. Hence $\text{dc} \leq 2k + 1$.

Now prove (31). Write $K = |J|$, and restrict P to the selected columns. If $K < 16$, the Gram matrix has diagonal entries one, so its norm is at least one and the result follows. Suppose $K \geq 16$ and let r_x be the number of selected lines containing point x . If $r_x < K/4$ for every x , take the unit vector $a = \mathbf{1}/\sqrt{K}$. At each point,

$$|\sum_{j \in J} a_j h_j(x)| = |2r_x - K|/\sqrt{K} > \sqrt{K}/2.$$

Its Gram Rayleigh quotient is at least $K/4$. Otherwise choose x_0 incident with $r \geq K/4$ selected lines and let S be those r lines. At x_0 all these labels are positive; at any other point at most one is positive, since two points have a unique common line. For the unit vector $a = \mathbf{1}_S/\sqrt{r}$,

$$|\sum_j a_j h_j(x)| \geq (r - 2)/\sqrt{r} \geq \sqrt{r}/2 \quad \text{for all } x,$$

since $r \geq 4$. Its Rayleigh quotient is at least $r/4 \geq K/16$, for every μ . This proves (31), including marginals concentrated on a single point.

The claimed learner queries $q_j(x, y) = y h_j(x)$ for every line and outputs a hypothesis with maximum answer. The true target has correlation one, hence receives answer at least $1 - \tau$. A maximizing hypothesis has true correlation at least $1 - 2\tau$, and loss at most τ , regardless of the marginal and the oracle. Thus it is one distribution-independent learner.

We prove the fixed- D claim before excluding a common efficient learner. Let $\delta = \epsilon/2$ and sample a length- s sequence from the known D . For any pair of hypotheses at D -distance greater than δ , the probability of identical labels on the sequence is at most $(1 - \delta)^s \leq e^{-\delta s}$. A union bound over fewer than N^2 pairs is at most $N^2 e^{-\delta s} \leq e^{-1} < 1$. Consequently some sequence has the property that hypotheses with identical traces are within D -distance δ . Hardwire one such sequence and one representative hypothesis per trace into the learner. On at most s distinct sampled points there are

at most $1 + s + \binom{s}{2}$ traces: the empty trace, at most s singleton traces, and at most one line through each pair of distinct sampled points. Every trace of size at least two is determined by one such pair. The representatives therefore form a D -cover of the claimed size. Query their label correlations and select the largest answer. A representative within δ of the true target has correlation at least $1 - 2\delta$. With tolerance $\tau = \epsilon/2$, the selected output has correlation at least $1 - 2\delta - 2\tau$ and error at most $\delta + \tau = \epsilon$. The hardwired cover depends on D ; this is not one learner serving all marginals.

To rule out a much cheaper common learner, choose for target line j the distribution

$$D_j(x) = \begin{cases} (2k)^{-1}, & P_{xj} = 1, \\ (2(N-k))^{-1}, & P_{xj} = 0. \end{cases}$$

Take the reference P_0 to be uniform on $X_q \times \{-1, +1\}$. With $c = k/(N-k)$, its residual likelihood ratios satisfy

$$r_j(x, +1) = A_{xj}, \quad r_j(x, -1) = -cA_{xj}.$$

For example the first formula is $2ND_j(x) - 1 = N/k - 1$ on the line and -1 elsewhere; the second is -1 on the line and $N/(N-k) - 1 = c$ elsewhere. Consequently

$$\begin{aligned} \Gamma &= \frac{1+c^2}{2N} A^\top A, \\ \Lambda &= \frac{qN}{2} \left(\frac{1}{(q+1)^2} + \frac{1}{q^4} \right), \\ R &= \frac{2}{q((q+1)^{-2} + q^{-4})} \geq \frac{(q+1)^2}{q} \geq q. \end{aligned}$$

The penultimate inequality uses $q^4 \geq (q+1)^2$ for $q \geq 2$, which follows from $q^2 \geq q+1$. By Theorem 2, $R \leq 4 + 4M$ for $M = m/\tau^2$. Therefore

$$\text{dc}(\mathcal{H}_q) \leq 2q + 3 \leq 2R + 3 \leq 11 + 8M \leq 19M,$$

where the final inequality uses $M \geq 1$. □

Why projective planes do not separate. The family has unbounded dimension while all the proposed ordinary label-Gram ratios remain bounded. This is a genuine failure of the stated universality premise, not a counterexample to OQ2. The efficient learner needed for such a separation is impossible here: the target-dependent balanced marginals produce a joint-distribution ratio of order q . The lower-bound witness changes entry magnitudes, which the ordinary label Gram matrix cannot capture. This also verifies an explicit separation between fixed-distribution and distribution-independent SQ learning: $m_D = O(n^2/\epsilon^2)$ at tolerance $\epsilon/2$, whereas for $q \geq 16$ any common learner needs $m/\tau^2 \geq q/4$. Since $N \leq 2q^2$, the latter is at least $2^{(n-1)/2}/4$. The dimension lower bound is at least $2^{(n-1)/4}$. None of these fixed- D learners can be substituted for the single algorithm in OQ2.

G.3 Transcript compression

Theorem 5 (Marginal-branching and approximate probabilistic bounds). *Let $B = \lceil 2/\tau \rceil + 1$, with $0 < \tau \leq 1$. For a deterministic learner satisfying (1), decompose each query as*

$$q(x, y) = a(x) + yb(x), \quad a(x) = \frac{1}{2}(q(x, +1) + q(x, -1)), \quad b(x) = \frac{1}{2}(q(x, +1) - q(x, -1)).$$

Construct its marginal-response tree as follows. If a is constant, retain only the answer equal to that constant. Otherwise retain the B equally spaced answers in $[-1, 1]$. Let u be the maximum number of nonconstant- a nodes on a retained root-to-leaf path. Then

$$\text{dc}(\mathcal{H}) \leq (m+1)B^u. \quad (33)$$

In particular, a deterministic correlational-only learner has $\text{dc}(\mathcal{H}) \leq m+1$. For an arbitrary randomized learner satisfying (1),

$$\text{dc}^\epsilon(\mathcal{H}) \leq B^m. \quad (34)$$

Proof. The grid has spacing $2/(B-1) \leq \tau$, so nearest-grid rounding errs by at most $\tau/2$. The marginal tree has at most B^u leaves: suppress unary nodes and use induction on the number of remaining branching nodes per path. It has at most $(m+1)B^u$ nodes, because every node has a descendant leaf and a depth between zero and m , and assigning each node one such pair is injective when the depth is fixed along a leaf's path. Collect the function b at every internal node and the Boolean output at every leaf. These are the common features $f_1, \dots, f_s$, where $s \leq (m+1)B^u$ and each feature has absolute value at most one.

Fix D, h . Follow the tree by giving the exact constant response or rounding $\mathbb{E}_D a$ to the grid. If at some node $|\mathbb{E}_D h b| > \tau/2$, one of the signed features $b, -b$ has correlation greater than $\tau/2$. Otherwise every answer is a valid SQ response: the difference from the true expectation $\mathbb{E}_D a + \mathbb{E}_D h b$ is at most $\tau/2 + \tau/2$. A valid policy realizing this finite transcript extends to all other histories by exact answers. The deterministic guarantee then gives leaf correlation at least $1 - 2\epsilon > 1/2 \geq \tau/2$. We have proved

$$\forall D, \quad \max_{i, s_i \in \{-1, +1\}} \mathbb{E}_D[h(x)s_i f_i(x)] \geq \tau/2. \quad (35)$$

We supply the finite separation step. Let C be the convex hull of the vectors $(h(x)s_i f_i(x))_{x \in X}$ and let $Q = (\tau/2)\mathbf{1} + \mathbb{R}_{\geq 0}^X$. If $C \cap Q$ were empty, the compact set C and closed set Q would have a positive attained minimum distance. To see attainment, take a minimizing sequence; the C components have a convergent subsequence, and bounded distances make the Q components bounded too. A zero minimum would put the limit in both sets. Let (c, z) minimize the positive distance and put $v = z - c$. By differentiating squared distance along line segments, for every $c' \in C, z' \in Q$ we obtain

$$\langle v, c' - c \rangle \leq 0, \quad \langle v, z' - z \rangle \geq 0.$$

Taking $z' = z + te_x$ gives $v_x \geq 0$. Furthermore $v \neq 0$, and

$$\max_{c' \in C} \langle v, c' \rangle \leq \langle v, c \rangle < \langle v, z \rangle = \min_{z' \in Q} \langle v, z' \rangle = (\tau/2) \sum_x v_x.$$

Normalizing v to a distribution contradicts (35). Hence a convex combination in $C \cap Q$ exists. Its coefficients define a linear combination of the common features whose product with $h(x)$ is at least $\tau/2$ at every point. This is a strict representation and proves (33). For correlational-only queries $a = 0$, hence $u = 0$.

For a randomized learner, fix a seed and instead retain all grid answers to each full query. There are at most B^m leaves. Use all leaf outputs as coordinates of an embedding, padding when needed. This distribution over embeddings depends only on the learner's seed, not on D, h . For fixed D, h , the policy rounding each true expectation to the grid is valid. Its output is one of the leaf features. Selecting that coordinate as w gives

$$\inf_w L_{D,h}(\text{sign}\langle w, \phi_\omega \rangle) \leq L_{D,h}(A_\omega^{O_{D,h}}).$$

Averaging over the seed proves (34). No seed is selected after observing D or h . $\square$

The deterministic argument does not derandomize the learner: fixing its seed need not preserve its expected guarantee. The probabilistic bound is exponential, not polynomial, and gives no exact $dc_{1/2}$ bound.

References

- [1] Vitaly Feldman, Pritish Kamath, and Nathan Srebro. Invited Open Problem: Is the Power of Deep Learning over Linear Models Inherently Distribution Dependent? *COLT*, PMLR 336:7117–7122, 2026. <https://proceedings.mlr.press/v336/feldman26a.html>.
- [2] Hamed Hatami, Pooya Hatami, William Pires, Ran Tao, and Rosie Zhao. Lower bound methods for sign-rank and their limitations. ECCC TR22-079, Sections 1–3 and Problem 4.1, 2022. <https://eccc.weizmann.ac.il/report/2022/079/>.
- [3] Noga Alon, János Pach, Rom Pinchasi, Radoš Radoičić, and Micha Sharir. Crossing patterns of semi-algebraic sets. *Journal of Combinatorial Theory, Series A*, 111(2):310–326, 2005. Theorem 1.3.
- [4] Hugh E. Warren. Lower bounds for approximation by nonlinear manifolds. *Transactions of the American Mathematical Society*, 133:167–178, 1968.
- [5] Noga Alon, Shay Moran, and Amir Yehudayoff. Sign rank versus VC dimension. *COLT*, PMLR 49:47–80, 2016. Full version arXiv:1503.07648, Theorem 21 and Lemma 22. <https://arxiv.org/abs/1503.07648>.
- [6] Vitaly Feldman. A general characterization of the statistical query complexity. *COLT*, PMLR 65:785–830, 2017. Definitions 4.6 and Theorems 4.8, 4.11. <https://proceedings.mlr.press/v65/feldman17c.html>.
- [7] Jürgen Forster. A linear lower bound on the unbounded error probabilistic communication complexity. *Journal of Computer and System Sciences*, 65(4):612–625, 2002.
- [8] Emmanuel Abbe, Pritish Kamath, Eran Malach, Colin Sandon, and Nathan Srebro. On the power of differentiable learning versus PAC and SQ learning. *NeurIPS* 34:24340–24351, 2021. <https://arxiv.org/abs/2108.04190>.
- [9] Hamed Hatami, Kaave Hosseini, and Xiang Meng. A Borsuk–Ulam lower bound for sign-rank and its applications. *STOC*, 463–471, 2023. ECCC TR22-130. <https://eccc.weizmann.ac.il/report/2022/130/>.
- [10] Mika Göös, Nathaniel Harms, Valentin Imbach, and Dmitry Sokolov. Sign-rank of k -Hamming distance is constant. arXiv:2506.12022v2, 2025. <https://arxiv.org/abs/2506.12022>.
- [11] Florian Frick, Kaave Hosseini, and Aliaksei Vasileuski. A $\mathbb{Z}_2$ -topological framework for sign-rank lower bounds. arXiv:2604.01510v2, 2026. <https://arxiv.org/abs/2604.01510>.
- [12] Ilias Diakonikolas, Christos Tzamos, and Daniel M. Kane. A Strongly Polynomial Algorithm for Approximate Forster Transforms and its Application to Halfspace Learning. arXiv:2212.03008, 2022. <https://arxiv.org/abs/2212.03008>.
- [13] Assia Mahboubi and Cyril Cohen. Formal proofs in real algebraic geometry: from ordered fields to quantifier elimination. arXiv:1201.3731, 2012. <https://arxiv.org/abs/1201.3731>.